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RESEARCH ARTICLE

GeoDisasterAINet: An Explainable Deep Ensemble Framework for Real-Time Urban and Rural Disaster Classification and Resilience

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ABSTRACT These Sustainable Development Goal 11 (SDG 11), whose target is to make urban and rural settlements resilient and disaster-prepared, integrates with it the significant objective of having disaster-resilient communities and cities. GeoDisasterAINet is an AI deep learning and machine learning model for distinguishing between rural and urban-area occurrences of floods, cyclones, earthquakes, and wildfires effectively. In this article, we introduce GeoDisasterAINet. There are four successive phases in model performance improvement, with increasingly enhancing classification accuracy, namely an integrated CNN (baseline model), an integrated CNN with XGBoost (feature improvement model), an integrated CNN with XGBoost and Multiclass SVM (refinement model for decision boundary), and LIME segmentation for interpretability and model performance evaluation. EfficientNetB7, ResNet-50, InceptionV3, and DenseNet-201 are architectures in use in GeoDisasterAINet, and these architectures make three settings for an ensemble: ERI-2025 (EfficientNetB7, ResNet-50, InceptionV3), DRI-2025 (DenseNet-201, ResNet-50, InceptionV3), and DE-2025 (DenseNet-201, EfficientNetB7). In balancing out the dataset, we used the Synthetic Minority Over-sampling Technique (SMOTE). With 99.25% accuracy in training, 96.37% accuracy in testing, 98% in precision, 97% in recall, and 97% in F1-score, ERI-2025 + XGBoost + Multiclass SVM, our best model, effectively classified items. LIME-based interpretability uncovered disaster-specific features such as storm behaviour in cyclones, infrastructure loss in earthquakes, wildfire propagation in

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wildfires, and water build-up in floods, and in confirming that the model was reliable, it helped in an effective location of disasters. GeoDisasterAINet is a strong AI-powered system that addresses directly SDG 11 through its contribution towards safer and disaster-resilient communities and cities. It does this through AI-powered deep learning and machine learning technology for improving disaster detection, evaluation of risk, and preparedness for emergencies in real-time.

INDEX TERMS SDG 11, deep learning ensembles, natural disaster classification, XGBoost and multiclass SVM, SMOTE for imbalance handling, LIME-based interpretability, GeoDisasterAINet framework.

I. INTRODUCTION

Natural disasters have catastrophic consequences for human life, infrastructure, and financial security, with growing intensity and frequency supported by environment degradation, urbanization, and climate change [1]. As per worldwide statistics, earthquakes, cyclones, floods, and fires have taken between 40,000 and 50,000 lives in a single year in the past two decades [2]. A record 363 high-impact weather events in 2023 in the world have touched over 93.1 million lives and have involved high financial loss. Global financial loss in terms of natural disasters in the whole world is \$3 trillion between 2010 and 2025, with developing countries taking most of the brunt with no proper infrastructure and preparedness mechanism in their hands. India, one of the most disaster-struck countries, alone experienced over 2,100 deaths in 2023, with most deaths being a consequence of floods, excessive rain, and heat waves. Cyclones between 1995 and 2020 have taken up 48% of deaths in India, with 26% of deaths having been a consequence of heat waves, 18% of deaths having been a consequence of floods, and 8% of deaths having been a consequence of cold waves [3]. Unfortunately, such statistics point towards urgency for developing high-tech disaster management programs with AI and deep learning tools for enhancing early warnings, minimizing reaction times, and enhancing efforts towards loss of property and life minimization [4].

Sustainable Development Goal 11 (SDG 11) of the United Nations seeks to develop safe, resilient, and sustainable urban and city communities [5], [6]. It puts a strong focus on urban infrastructure development, effective disaster management frameworks, and quick post-disaster restoration processes in case of any extreme weather occurrences. The logic has been long embedded, as urban communities devastated by natural disasters have paid a heavy price over the decade, both in terms of high density, poor urban planning and poor drainage infrastructure. Rural communities on the other hand continue to face higher vulnerability to fires and flooding, entailing high environmental degradation impact and loss of livelihoods, in most cases, there was low preparedness and an ineffective response mechanism in place. In India, urbanization and a changing environment have also increased the vulnerability to extreme weather, with cyclones and flooding displacing a large section of its population, causing heavy loss to India's key socio-economic infrastructure. A report by NDMA [7], India: 500 million people were displaced between 2010 and 2025, and the economic loss is estimated

to be more than XXXX7 lakh crore (85 billion USD) [8], [9]. Improved policies represent a huge progress, but predictive analysis, and real time disaster reaction have some gaps in it. AI/ML powered solutions can fill such gaps and help to make the communities disaster resilient [10].

Artificial intelligence (AI) platforms for real-time event classification, risk analysis, and early warnings have become a breakthrough in disaster management in consideration of such concerns [11]. Conventional approaches to discovering and managing disasters depend almost exclusively on past data and satellite observation, and cannot necessarily early warn about rapid onset events in terms of climate [12]. Recent advancements in deep learning, machine learning, and ensemble modelling have yielded substantial enhancements in the classification of natural disasters, the assessment of their impacts, and the planning for recovery [13]. Combining Convolutional Neural Networks (CNNs), XGBoost, and Multiclass Support Vector Machines (SVMs) has shown to be very effective at making disaster classification more accurate [14]. Also, techniques like the Synthetic Minority Over-sampling Technique (SMOTE) make it easier to make balanced datasets, which makes training processes more reliable [15]. Additionally, Local Interpretable Model-Agnostic Explanations (LIME) help make models easier to understand, which improves the reliability and transparency of AI-based disaster management systems [16].

To achieve target 11 of SDG, in safer and more resilient communities and cities, artificial intelligence (AI), deep learning (DL), and machine learning (ML) are utilized in GeoDisasterAINet for rapid and accurate catastrophe detection. National and international disaster management departments can use such a system, capable of early disaster detection, providing real-time information, and ease in assigning resources [17]. With a growing number of climate-related disasters, an AI-facilitated, scalable disaster management system is increasingly significant. In this work, a robust platform for artificial intelligence-facilitated disaster resilience is developed, and deep learning ensembles, for example, can revolutionize disaster preparedness, response, and rehabilitation for urban and rural communities worldwide.

This paper presents GeoDisasterAINet, a framework for deep learning ensembles that uses artificial intelligence (AI) to classify natural disasters in both urban and rural locations. Its goal is to help achieve SDG 11, which is to become more resilient to disasters. Integrated Convolutional

Neural Networks (CNNs) in four stages are employed in the model: Integrated XGBoost with CNNs for basic classification, Integrated XGBoost with CNNs and Multiclass Support Vector Machines (SVM) for optimized classification, and LIME-based visualization for model simplification and testing [18]. The utilized dataset consists of four significant types of disasters: cyclones, earthquakes, floods, and wildfires. It was trained with deep architectures of convolutional neural networks, i.e., EfficientNetB7, ResNet-50, InceptionV3, and DenseNet-201. With ERI-2025, XGBoost, and Multiclass SVM, the best model attained an F1 value of 97%, a value for precision at 98%, a value for recall at 97%, and a value for accuracy at 96.37%. It is a much better performance in comparison with traditional disaster classification models. Real-time analysis and prediction, through this AI model, allow policymakers, first respondents, and urban planners to form effective disaster-mitigation strategies.

II. LITERATURE SURVEY

Using Artificial Intelligence (AI), Machine Learning (ML), and Deep Learning (DL) in disaster management has changed how we detect, evaluate, and mitigate disaster impact. Traditional approaches to disaster classification rely on satellite, social media, and sensor information. These require a lot of work and a long processing time. Scholars have developed automated and scalable techniques for disaster classification through complex deep learning architectures, multi-modal fusion approaches, and group approaches. These advances contribute towards SDG 11, whose target is to make urban and rural areas disaster resilient.

Recent research has considered numerous AI techniques for disaster detection. It entails discovering disasters through social media, satellite picture analysis, victim identification, and post-disaster analysis. Text and multimodal techniques with Convolutional Neural Networks (CNNs), Long Short-Term Memory networks (LSTMs), and Transformer architectures function effectively in gathering information regarding disasters through social media platforms such as Twitter. It keeps humans informed about disasters in real-time. Nevertheless, such techniques suffer from disparate language usage, unorganized information, and poor contextual awareness. Application of picture-based disaster detection with Convolutional Neural Networks (CNNs), Generative Adversarial Networks (GANs), and Vision Transformers have proven useful. They can tag ruined buildings, detect affected areas, and assess impact post-disaster. Challenges in such a system include its dependability of information, high computational expense, and generalizability over a variety of disaster scenarios.

The accuracy of catastrophe classification has also been enhanced by the emergence of hybrid AI models that combine deep learning with machine learning. Many studies have used ensemble learning methods that combine Convolutional Neural Networks (CNNs) with boosting techniques like XGBoost, Gradient Boosting, and Multiclass Support Vector

Machines (SVM). These hybrid techniques apply both feature extraction and decision improvement, and such approaches make disaster classification frameworks more accurate. Most of the studies have no explicit explanations, and it is not an easy matter to trust AI classifiers in real-life scenarios for disaster classification. Methods such as LIME (Local Interpretable Model-Agnostic Explanations) and SHAP (SHapley Additive Explanations) have become increasingly prevalent in model transparency improvement.

This literature review in Table 1 compares and contrasts emerging AI disaster classification models in terms of function, accuracy, strengths, and weaknesses. It identifies gaps in ongoing disaster classification-related studies and proposes a new powerful and readable deep model, GeoDisasterAINet, to fill such gaps.

A. RESEARCH GAP AND NEED TO WORK ON GeoDisasterAINet

A thorough review of current studies in AI-based disaster classification reveals significant gaps in studies that make them less effective and applicable in real-life scenarios. One such significant constraint is that the most effective deep neural networks, including CNNs for disaster classification and Vision Transformers, lack interpretability and transparency. They don't use tools such as LIME and SHAP that can interpret their behaviour. Because of a lack of interpretability, they become less readable in critical decision-making scenarios. Most disaster classification models only attend to satellite images alone, social media events, or both but not both together for a complete analysis. Consequently, most of them excel in a single type of disaster, and therefore, it's not easy for them to work effectively in a range of scenarios. Real-time tracking of a disaster is a big challenge yet to overcome. Most studies represent that it can become remarkably accurate, but in practice, it fails in most scenarios because it takes a lot of computational power and doesn't scale well. This makes it hard to use in emergency response systems. Also, having unbalanced data and biased models greatly weakens the trustworthiness of classification results. This is mainly because most models rely on data that has been manually labelled, which leads to poor performance and mistakes in real-world situations. Only a few studies have used the Synthetic Minority Over-sampling Technique (SMOTE) or other methods to improve the variety and fairness of datasets. In conclusion, combining deep learning with advanced machine learning methods like XGBoost and multiclass support vector machines is not often used, even though it can improve classification results. To address such issues, GeoDisasterAINet couples deep learning with multi-facets of artificial intelligence, real-time observation, utilizes SMOTE for balancing, and LIME for improving its result interpretability. All these in integration enable a transparent, expandable, and efficient mechanism for disaster classification, in support of Sustainable Development Goal 11, to improve disaster management in urban and rural environments.

TABLE 1. Literature survey.

Authors & Year	Disaster Type	Methodologies Utilized	Accuracy Obtained	Advantages	Limitations	Research Gap	Real-Time Feasibility
Yu, C., Hu, B., & Wang, Z. (2024) [19]	Open-world disaster event detection	Multimodal deep learning model integrating textual and image-based recognition	92%	Real-time multimodal detection, early warning	High computational resources, model biases	Reduced computational complexity via optimized ensemble methods	Medium
Singh, R., & Sharma, K. (2023) [20]	Social media disaster prediction	CNN, LSTM, and Transformer-based tweet analysis	89%	Supports fast emergency response	Limited by informal texts and language variations	Visual-based classification avoiding NLP limitations	High
Seeja, G., Selvakumar, A. A., & Hency, V. B. (2023) [21]	Victim detection	ResNet-50 and Decision Tree for detecting victims under debris	99.53%	High accuracy, quick rescue operations	Limited generalization (body part-specific)	Broader disaster generalization and interpretability	Medium
Gupta, A., & Banerjee, P. (2023) [22]	Natural disaster classification	CNN models on satellite imagery	90%	High accuracy, precise response planning	Extensive labeled data, limited low-res images	Effective with fewer labeled samples via SMOTE balancing	Medium
Islam, M. A., Rabbi, F., & Hossain, N. U. I. (2024) [23]	Social media disaster identification	NLP (BERT, CNN) for textual classification	94%	High accuracy, efficient training	Handling noisy and ambiguous texts	Visual data analysis avoiding textual ambiguity	High
Mali M. et al. (2024) [24]	Disaster tweet classification	Comparative analysis (LR, Naïve Bayes, CNN, BERT)	Not specified	Foundational analysis of ML/DL models	Lacks empirical validation	Empirically validated visual classification models	Medium
N. Shivakumar (2022) [25]	Infrastructure damage assessment	GAN-based data augmentation with CNN	93%	Fast, reduces manual assessment	Poor performance in adverse conditions	Robust model performance across diverse conditions	Medium
Irene Alisjahbana et al. (2024) [26]	Building damage from satellite images	U-Net segmentation and CNN classification	95%	Accurate segmentation and classification	Computationally intensive, slow training	Optimized real-time deployment capabilities	Medium
Robinson Umeike et al. (2024) [27]	Tornado damage classification	Vision Transformer for post-disaster assessment	96%	High accuracy, rapid assessment	Limited generalizability beyond tornadoes	Broad applicability across diverse disaster types	Low
Asim Bashir Khajwal et al. (2022) [28]	Multi-view building damage classification	Multi-view CNN classification	91%	Improved accuracy from multiple perspectives	Reduced accuracy with occlusions	Robust to occlusions and generalization	Low
Branislava Jankovic et al. (2025) [29]	UAV-based real-time disaster detection	Optimized Transformer model with UAV	94.50%	Scalable, real-time UAV application	Edge computation constraints, UAV-specific limitations	Ground-based high accuracy and real-time capability	High
Md Safwan Mondal et al. (2025) [30]	UAV-UGV disaster surveillance	DRL framework using transformer architectures	Not specified	Persistent surveillance, optimal routing	High computational load, complex implementation	Simplified real-time disaster classification	High
Sayan Kumar Bhowmick et al. (2025) [31]	Explainable AI models evaluation	Threshold metrics for evaluating interpretability	Not specified	Improved interpretability and trustworthiness	Focuses on metrics without broader disaster context	Practical integration of LIME interpretability in disaster scenarios	Medium

B. TECHNOLOGIES UTILIZED IN GeoDisasterAINet

GeoDisasterAINet is a smart AI tool for deep learning and machine learning to classify, identify, and react to disasters in real time. It integrates the most advanced AI techniques, specifically Convolutional Neural Networks (CNNs), XGBoost, and Multiclass SVMs. In such a system, the classification of disaster is done by CNN for extracting features, as XGBoost is for feature selection and Multiclass SVM is for the maximization of its decision. By using such a classification mechanism, the system specifies types of disasters, such as cyclones, earthquakes, floods, and fires, including wildfires, and then it detects them not only in urban areas but also in rural areas. To solve the problem of imbalanced datasets, GeoDisasterAINet uses the Synthetic Minority Over-sampling Technique (SMOTE), a method for additional sample generation to balance datasets in the feature space by synthetically producing more samples based on under-class samples. This approach also enhances the model's generalizability across various disaster contexts. We also implement LIME (Local Interpretable Model-Agnostic Explanations) to enhance the transparency and interpretability of our model. Such a feature allows for disaster classification to be comprehended and validated by way of indicating important areas in images for decision-making to groups during the incident response process. Additionally, GeoDisasterAINet is the first to detect disasters in real time and offers a feature that allows for flexible, efficient, and effective emergency response in both urban and rural regions. It enables situational awareness, real-time disaster detection, and efficient asset utilization if such a feature exists. GeoDisasterAINet equipped with state-of-the-art artificial intelligence integration means that it completes Sustainable Development Goal 11 (SDG 11), a key goal for making urban and rural communities resilient to disasters and sustainable. This paves the way for the development of an innovative AI-driven global disaster management platform.

III. METHODOLOGY

GeoDisasterAINet is an ensemble deep learning model that utilizes a mix of AI approaches for real-time analysis and classification of disaster impacts [32]. The model works with a four-step mechanism with a target of improving classification accuracy incrementally with the intention towards offering scalability in frameworks for disaster response. All three phases include in-depth feature extraction and classification, with model interpretability improvement in the last stage. All three complex CNN architectures, namely ERI-2025, DRI-2025, and DE-2025, are utilized in the model. ERI-2025 incorporates EfficientNetB7, ResNet-50, and InceptionV3; DRI-2025 incorporates DenseNet-201, ResNet-50, and InceptionV3; and DE-2025 incorporates both DenseNet-201 and EfficientNetB7. All three have been trained with an incorporation of the use of the Natural Disaster Dataset, with the four most important types of disasters: cyclones, earthquakes, flooding, and wildfires. All three

CNNs in combination make up the default configuration for feature extraction, with feature extraction improvement in later phases of GeoDisasterAINet.

Figure 1 shows the four steps of GeoDisasterAINet. In the first step, it uses combined CNNs (ERI-2025, DRI-2025, and DE-2025) to classify disasters. The second stage enhances classification by combining Convolutional Neural Networks (CNNs) with XGBoost, which helps in better selecting important features. The third stage sharpens the boundary of the ultimate decision with Multiclass Support Vector Machines (SVMs), to improve accuracy in disaster classification. LIME (Local Interpretable Model-Agnostic Explanations) in stage four makes the decision of the system transparent. It helps the system in finding and highlighting important regions in an image contributing to its classification output. In addition, the application of the Synthetic Minority Over-sampling Technique (SMOTE) corrects any problem with class imbalance, a prevalent problem with real-life disaster datasets, and aids in balancing a dataset. GeoDisasterAINet is a flexible and real-time system that helps in discovering disasters. It is a dependable AI system supporting Sustainable Development Goal 11 (SDG 11), whose target is improving disaster preparedness in urban and rural areas.

A. DATASET DESCRIPTION AND PREPROCESSING

This research is predicated on a dataset pertaining to natural disasters, which comprises unprocessed images depicting four principal categories of disasters: cyclone, earthquake, flood, and wildfire [33]. The dataset exhibited a considerable disparity in its original configuration, with certain disaster categories comprising a substantially greater number of images compared to others. To tackle such an issue, the dataset was split into three sets: training, validation, and testing sets, in a 75%, 15%, and 10%, respectively, proportionate distribution. This helped in providing an open and transparent way in which to evaluate GeoDisasterAINet. The dataset used in this study was initially composed of 4,428 images that were gathered from publicly available sources. The images were specifically categorized into four main categories of natural disasters: cyclones (928 images), earthquakes (1,350 images), floods (1,073 images), and wildfires (1,077 images). The dataset was purposefully separated into three subsets: training (3,322 images, approximately 75%), test (662 images, approximately 15%), and validation (444 images, approximately 10%) for systematic assessment. The original images were subjected to an array of operations for preprocessing since all images were in different resolutions and formats. To satisfy the input requirements of deep learning architectures—EfficientNetB7, ResNet-50, InceptionV3, and DenseNet-201—all images were converted to 224×224 pixels uniformly. Besides, image preprocessing applied Gaussian noise elimination to remove extraneous artifacts, min-max normalization to normalize pixel values to lie between 0 to 1 uniformly, and standardized color conversion to RGB to ensure that all CNN architectures were

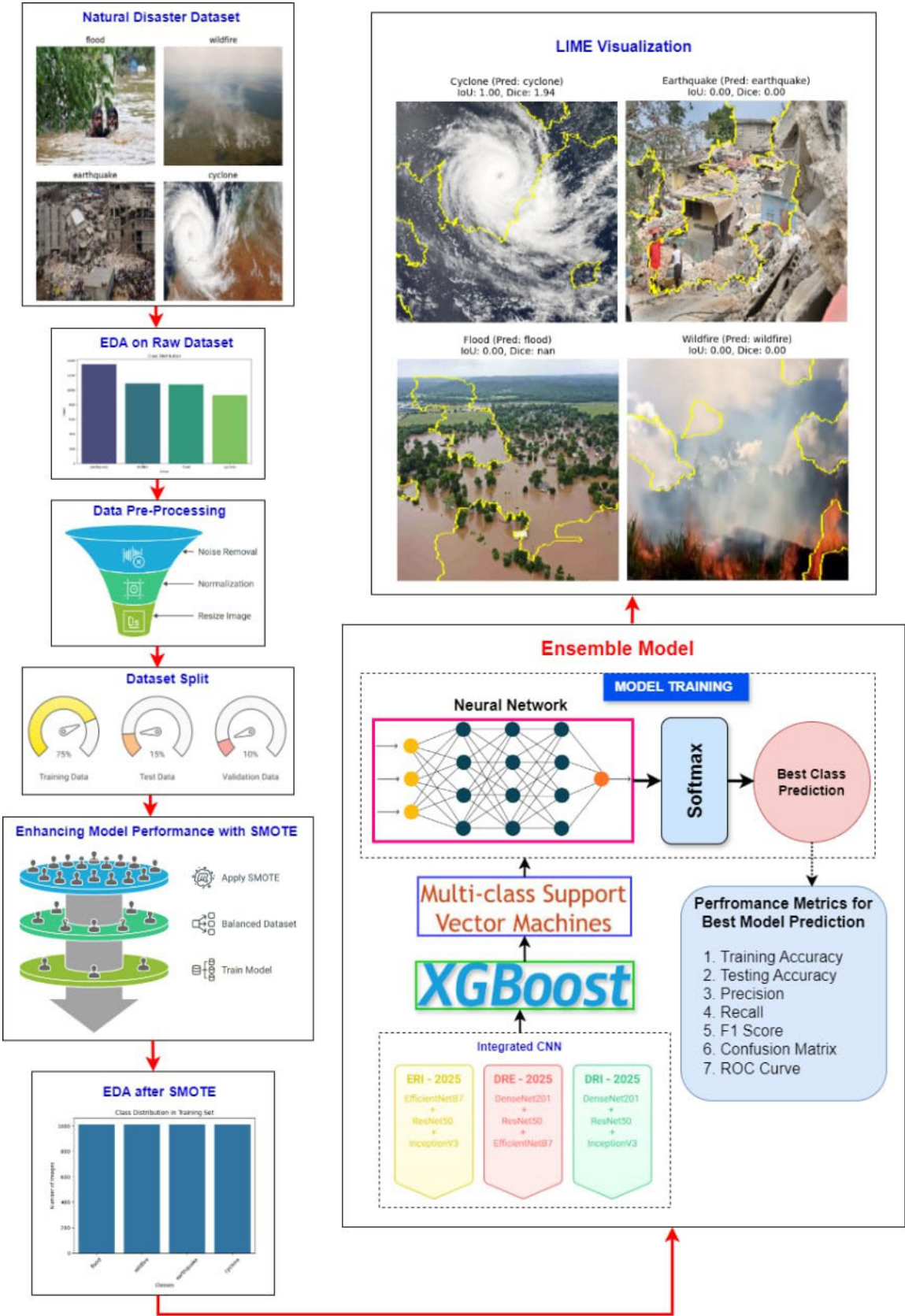


FIGURE 1. The ensemble GeoDisasterAINet framework.

compatible with it. In order to address the class imbalance that was observed in the original dataset, particularly the under-representation of cyclone images (only 696 training images initially compared to 1,013 earthquake images), we implemented the Synthetic Minority Over-sampling Technique (SMOTE) on the training subset. The dataset was effectively balanced by SMOTE, which increased the representation to 1,012 images per disaster type (cyclone, earthquake, flood, and wildfire). This resulted in a balanced training set comprising 4,048 images. However, the testing and validation sets remained unchanged to maintain realistic evaluation conditions. Figure 2 shows the distribution of classes, and an issue with a problem of a class imbalance, and such a problem must therefore be resolved in preparation for model training. There was a high presence of earthquakes, with cyclones having a minimum presence. It is therefore imperative to use techniques for balancing data in an attempt to make training performance bias free.

We performed an Exploratory Data Analysis (EDA) of this raw data in an attempt to understand picture properties and distribution over classes more deeply [34]. Figure 3 shows a selection of images in the dataset and reflects the diversity and complexity of disaster photographs. They represent a range of environments, levels, and locations of disasters. Exploratory data analysis (EDA) helped in checking for picture quality, resolution invariability, and the presence of any potential noise in the dataset. EDA aided preprocessing and augmentation processes, a requirement for model performance enhancement. By overcoming such complications in the data, GeoDisasterAINet enables fair and efficient disaster classification. This strengthens its role as an AI system for Sustainable Development Goal 11 (SDG 11), whose objective is to build urban and rural community disaster resilience.

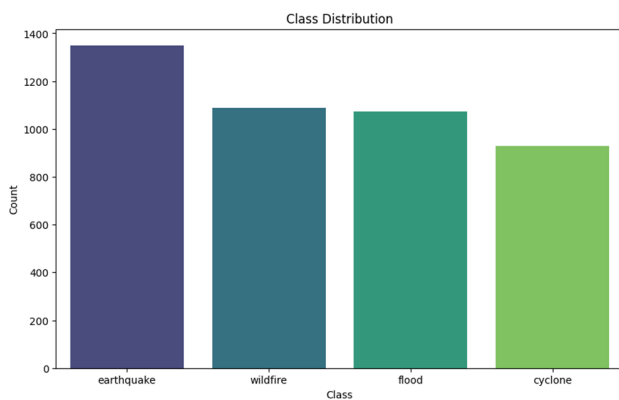


FIGURE 2. The ensemble GeoDisasterAINet framework.

Table 2 describes the raw image dataset utilized in this work. It separates the images into training, testing, and validation sets for four categories of disasters: cyclone, earthquake, flood, and wildfire. Initially, the dataset was characterized by an imbalance, with earthquake images representing the most dominant group, with 1,350 images, and cyclone images representing the least, with 928 images. We split the dataset

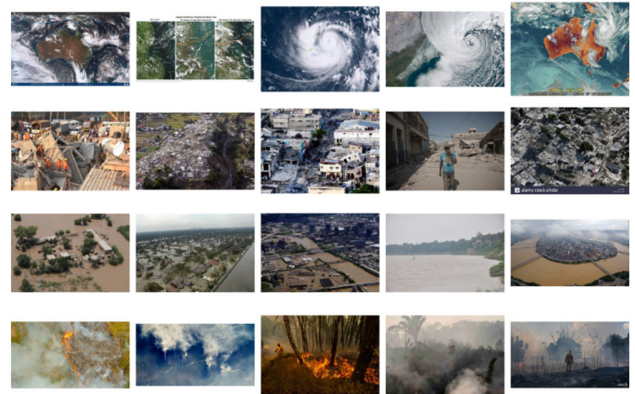


FIGURE 3. The natural disaster dataset sample images.

into 75% for training, 15% for testing, and 10% for validation for an orderly evaluation process. This split helps in efficient model training and an objective evaluation of performance in GeoDisasterAINet.

B. PREPROCESSING OF RAW DATASET IMAGES

The raw data was cleaned and prepared in a manner of minimizing noise, normalizing, and re-sizing the images to 224×224 pixels to make the input images for GeoDisasterAINet reliable and uniform [35]. Noise removal was conducted to eliminate unnecessary distortions and complications that could have a deleterious impact on feature extraction in deep learning algorithms. Normalization was conducted to normalize between 0 and 1 for efficient training of the convolutional neural network (CNN) architectures and to avert any bias towards specific intensity values. All the images have to have been re-sized to 224×224 to have a uniform input for a range of convolutional neural network (CNN) architectures including EfficientNetB7, ResNet-50, InceptionV3, and DenseNet-201. It aids in compatibility and computational efficiency in terms of standardization. Preprocessing is an important aspect of computer vision operations. It simplifies visual features, reduces processing, and enables a model to function effectively with a range of types of information. In case of poor preparation, deep learning algorithms will become confused with noise and important patterns, and poor classification accuracy will occur. GeoDisasterAINet improves disaster detection by making the data consistent and removing unnecessary differences. This helps the system find important patterns more effectively, speeds up its learning process, and makes disaster classification more accurate, which is crucial for responding to emergencies in real-time.

C. PREPROCESSING OF RAW DATASET IMAGES

A big challenge in training deep learning models for classifying disasters is the unevenness in the dataset. This happens when some types of disasters have too many examples while others have very few. In GeoDisasterAINet, the imbalance was obvious in the raw dataset, where there were many

TABLE 2. The raw dataset distribution.

Dataset Split	Cyclone	Earthquake	Flood	Wildfire	Total	Image Resolution	Preprocessing Steps	Data Balancing
Training	696	1,013	805	808	3,322	224 × 224 pixels	Noise removal, normalization, resizing	SMOTE applied (balanced to 1,012 images per class)
Testing	139	201	160	162	662	224 × 224 pixels	Noise removal, normalization, resizing	No balancing applied
Validation	93	136	108	107	444	224 × 224 pixels	Noise removal, normalization, resizing	No balancing applied
Total	928	1,350	1,073	1,077	4,428	Uniform Resolution	Standardized preprocessing for consistency	Balanced Training Set

images of earthquakes but very few images of cyclones, as shown in Figure 2. Such imbalances result in unbalanced model training, and the classifier will preferentially orient towards the dominant class. This generates less generalization capabilities and less accuracy in predicting less common types of disasters. To counteract such a problem and have balanced training, we have adopted a method called the (SMOTE) for application with the dataset [36], [37].

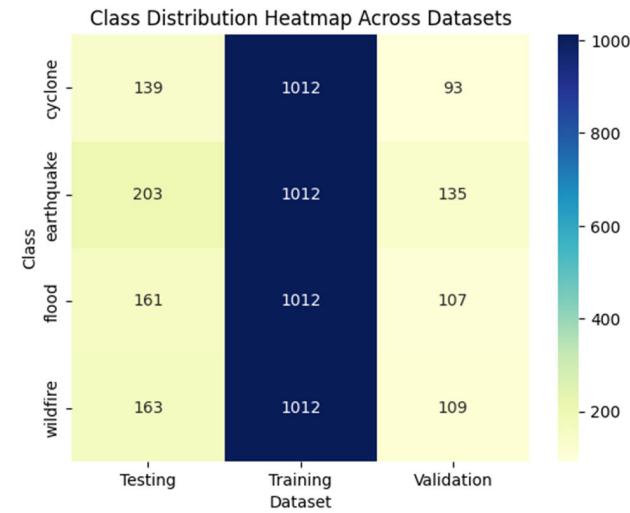


FIGURE 4. SMOTE balancing of the dataset.

SMOTE creates synthetic samples for minority classes instead of copying existing images. Doing this, reinforces the model with assured diversity in disaster types, for example, cyclones and floods, during training. It minimizes bias and optimizes classification accuracy for all disaster types. After using SMOTE on the training dataset, we did more Exploratory Data Analysis (EDA). This included looking at the distribution of the balanced dataset and showing sample images, as seen in Figure 4 and Figure 5. The dataset was then resized to 224 × 224 pixels to assure consistency across CNN architectures. After using the Synthetic Minority

Over-sampling Technique (SMOTE), the training dataset now has an equal number of images in the categories of cyclones, earthquakes, floods, and wildfires, with each category containing 1,012 images. Figure 4 depicts such distribution in a distribution of classes in the form of a heatmap. Having a balanced dataset is important in training deep algorithms, and it helps in bias minimization, accuracy in classification, and fair identification of any form of disaster.

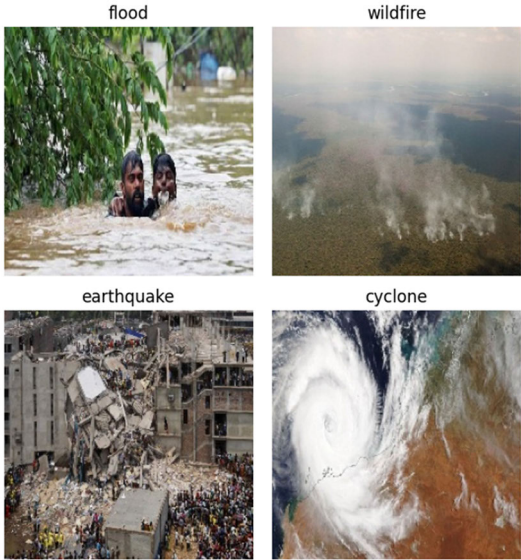


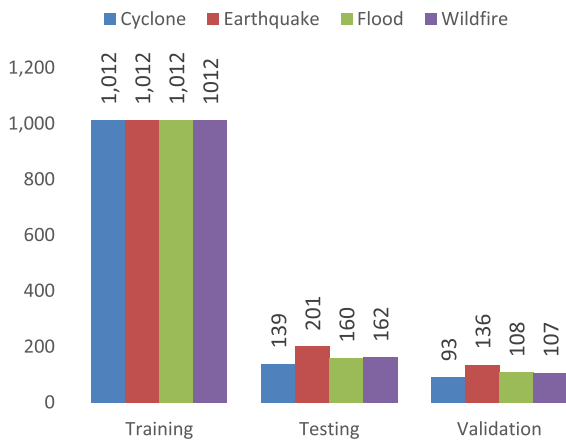
FIGURE 5. After Pre-processing and SMOTE sample images.

This balanced information is significant in training Convolutional Neural Networks (CNNs), XGBoost, and Multiclass Support Vector Machines (SVMs) in the GeoDisasterAINet model. It aids in developing a balanced training environment, reducing room for mistakes in classification and improving the model’s ability to apply its gained information. By using SMOTE, feature extraction with CNN yields a balanced view of the features. XGBoost effectively identifies key features, and Multiclass SVM effectively distinguishes between types of disasters. Having a balanced dataset greatly enhances the

TABLE 3. After SMOTE balancing dataset distribution.

Dataset Split	Cyclone	Earthquake	Flood	Wildfire	Total
Training	1,012	1,012	1,012	1012	4,048
Testing	139	201	160	162	662
Validation	93	136	108	107	444
Total	928	1,350	1,073	1,077	5,154

capability of the framework to classify disasters in real-time. This makes GeoDisasterAINet a strong and adaptable AI tool that addresses Sustainable Development Goal 11 (SDG 11), which addresses disaster resilience. Table 3 and Figure 6 illustrate the dataset that has been balanced using the SMOTE (Synthetic Minority Over-sampling Technique) methodology.

**FIGURE 6. The distribution of train, test and validation after SMOTE.**

D. MODEL TRAINING IN GeoDisasterAINet FRAMEWORK

1) CNN MODELS UTILIZED IN GeoDisasterAINet FRAMEWORK

GeoDisasterAINet uses four high-performance deep learning models: EfficientNetB7, ResNet-50, InceptionV3, and DenseNet-201. All four contribute in a specific way towards disaster classification. EfficientNetB7 is a well-engineered model of a convolutional neural network (CNN) that utilizes a compound scaling technique [38]. This effectively scales both the model's resolution, width, and depth in a balanced manner. With this, the model can obtain high accuracy and utilize fewer parameters compared to conventional CNNs [35]. With such a feature, it is particularly useful for big datasets of calamity, in which computational efficiency is a key issue. ResNet-50, a 50-layered residual network, is engineered with skip connections, alternatively known as residual connections, that enable the model to bypass specific layers [39]. With such an architectural feature, vanishing gradients, a problem in deep networks in general, are effectively countered. With its feature extraction capabilities, specifically in detail, ResNet-50 is best for discovering structures in

a disaster such as a collapsed building, a flood, and a wildfire, for example. In contrast, InceptionV3 is particularly useful in that it carries multi-scale convolutional layers that enable it to garner features at a range of scales [39]. With such a feature, the network can detect big and small disasters, such as a big flood and a fissure in a building during an earthquake, respectively. DenseNet-201 is a 201-layered neural network that utilizes a new mechanism in that it connects each layer with any other layer [41]. With such a configuration, it maximizes feature use and reduces redundancy, creating a model that can effectively garner both simple and complex information about a disaster scenario.

The three bespoke ensemble models, ERI-2025, DRI-2025, and DE-2025, along with their corresponding specialized CNN models, form a multi-stage ensemble model approach for an effective disaster classification system [42]. To analyse complex disaster trends combined with a range of over-ground and satellite images to make predictions, ERI-2025 deploys an EfficientNetB7, a ResNet-50, and an InceptionV3. ERI-2025 utilizes EfficientNet's depth-wise scaling, deep learning capabilities of ResNet, and Inception feature extraction at a range of scales. ERI-2025 is particularly effective in detecting such disasters as fires in thick forests, tsunamis, and hurricanes. DRI-2025 incorporates three model strengths: DenseNet, ResNet, and InceptionV3. DRI-2025 utilizes multi-scale InceptionV3 feature extraction, deep learning in ResNet, and feature sharing in DenseNet. DRI-2025 is effective in distinguishing earthquakes' impacts, such as scarred lands and collapsed buildings. High connectivity in DenseNet helps in distinguishing minor trends, such as precursors to lands shifts and destroyed buildings. It helps in enhancing preparedness and lessening the impact of disasters.

The DE-2025 model, a DenseNet-201 and EfficientNetB7 combination, is particularly optimized for efficient classification at a low computational cost, with rapid real-time disaster identification capabilities. DenseNet can effectively reuse its features, and with EfficientNet's efficient scaling, such a model is particularly effective at detecting rapidly emerging disasters such as wildfires, destructive floods, and cyclones. Figure 7, Figure 8, Figure 9, Figure 10, Figure 11, Figure 12, and Figure 13 illustrate in detail these architectures, explaining in detail how their layers are interconnected, blocks of convolution, and how they work together in the GeoDisasterAINet model. With such group convolutional neural network (CNN) architectures, GeoDisasterAINet performs best in accuracy, reflects increased capability in its adaptability in terms of types of disasters, and provides dependable output when classifying in real-time, distinguishing between types of disasters. It is a strong AI tool for emergency response and Sustainable Development Goal 11 (SDG 11) programs.

2) XGBOOST FOR GeoDisasterAINet FRAMEWORK

XGBoost is an efficient machine learning algorithm. It is utilized in GeoDisasterAINet for enhancing disaster classification accuracy through gradient-boosted decision trees

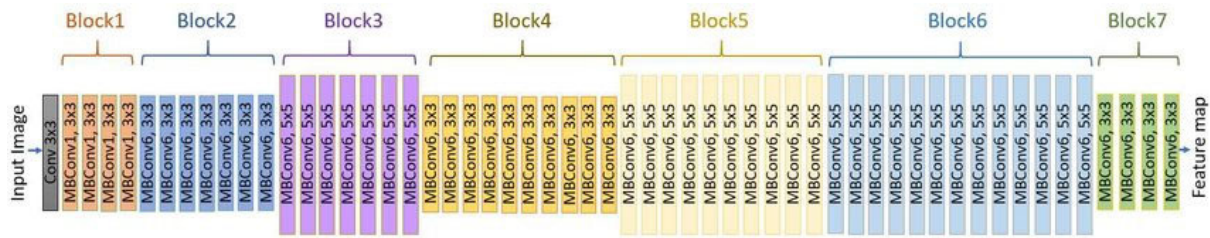


FIGURE 7. The architecture of CNN: EfficientNetB7.

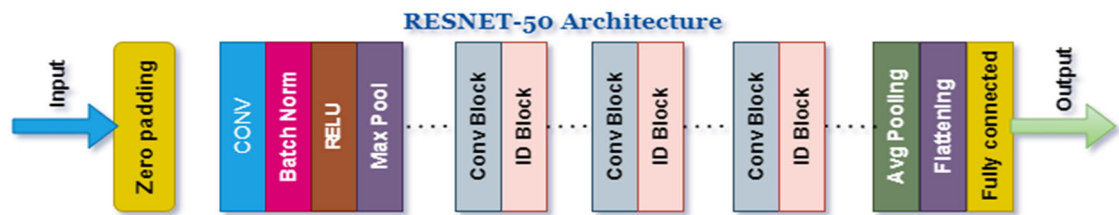


FIGURE 8. The architecture of CNN: ResNet50.

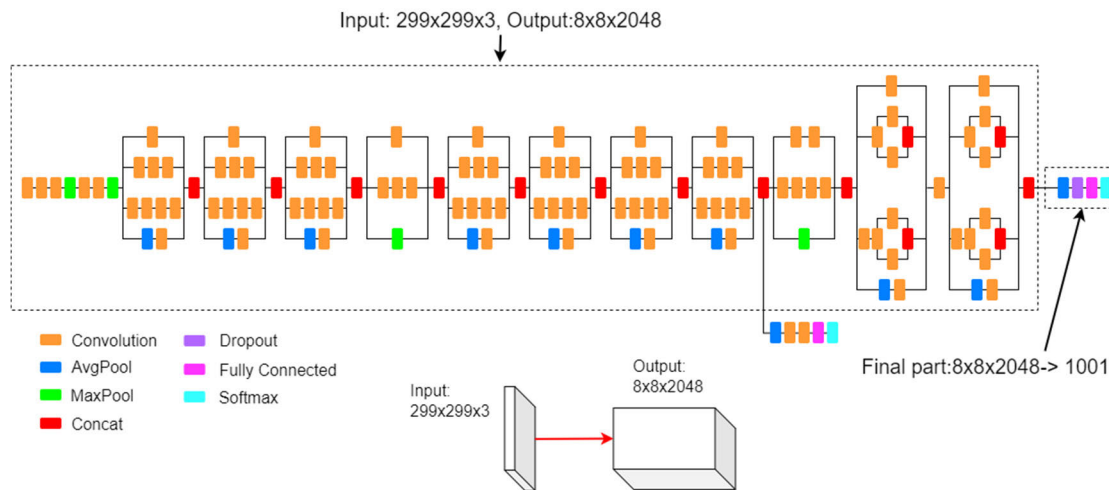


FIGURE 9. The architecture of CNN: InceptionV3.

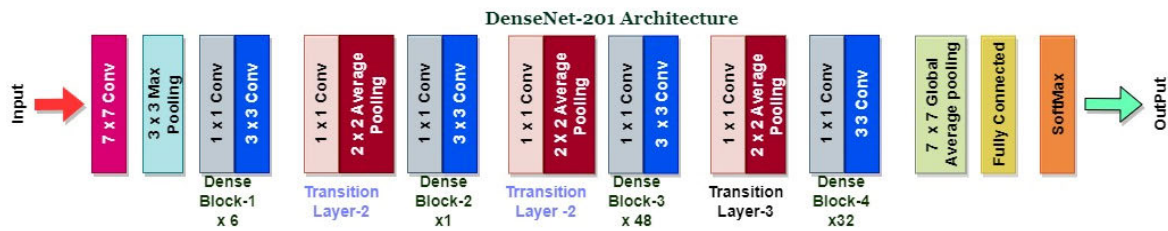


FIGURE 10. The architecture of CNN: DenseNet-201.

(GBDTs). Figure 14 depicts that XGBoost keeps updating bad decision trees, and an individual tree corrects the errors of its own [43], [44]. Due to its iterative boosting, the model

puts a growing concern towards misclassified disaster samples, sharpening its cyclone, earthquake, flood, and wildfire classification decision. Decision Tree 1 first comes with an

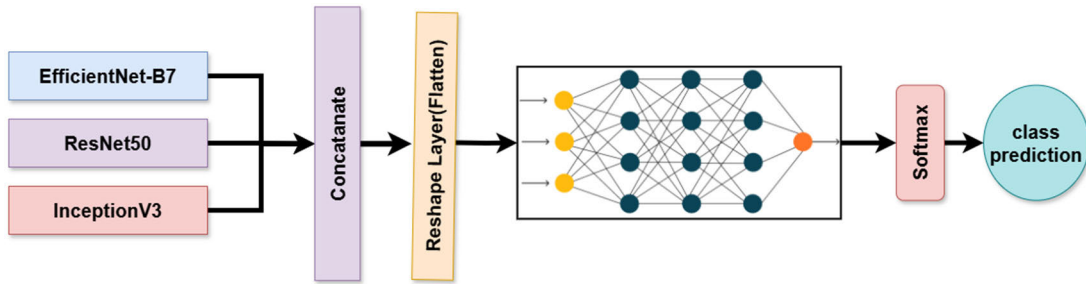


FIGURE 11. The architecture of integrated CNN: ERI-2025.

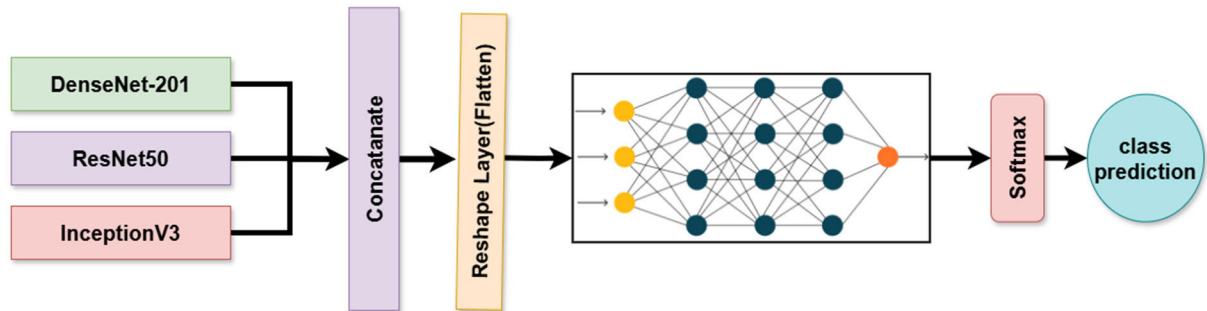


FIGURE 12. The architecture of integrated CNN: DRI-2025.

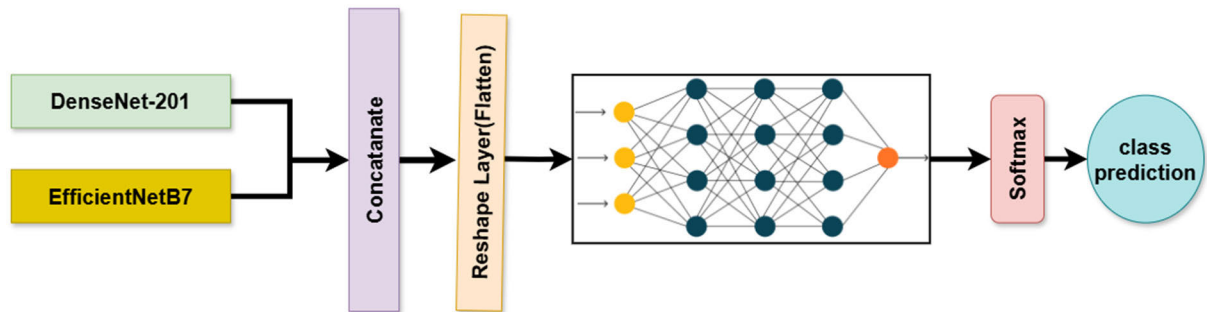


FIGURE 13. The architecture of integrated CNN: DE-2025.

initial classification through extracted features in Integrated CNNs (ERI-2025, DRI-2025, and DE-2025). All successive trees are correct for wrongly classified samples, offering a refinement in terms of disaster determination in a better manner.

Upon producing multiple decision trees, XGBoost uses a weighted aggregation strategy, where each tree's output is combined depending on its contribution to the total accuracy of classification [45]. This ensemble strategy increases the power of feature selection, and therefore, XGBoost is highly effective in distinguishing disaster patterns learned from features obtained by convolutional neural networks (CNNs). The final step is the application of the SoftMax function, which transforms the combined outputs into probability scores for class predictions, enabling correct classification of real-world disaster cases. The use of XGBoost in GeoDisasterAINet

greatly improves classification performance in comparison to individual deep learning models, and it aids in improving resilience to class imbalance as well as enabling improved disaster detection in real-time applications.

By taking advantage of the strength of XGBoost in handling noisy and high-dimensional data, GeoDisasterAINet ensures best-in-class decision-making in complex disaster scenarios and hence enables enhanced disaster classification and response planning. The boosting technique enhances feature learning, especially for small disaster classes, like less documented cyclone events, and hence the accuracy of the system in actual emergency scenarios. This integration achieves Sustainable Development Goal 11 of disaster resilience through the delivery of a highly scalable and interpretable AI-driven model for disaster classification.

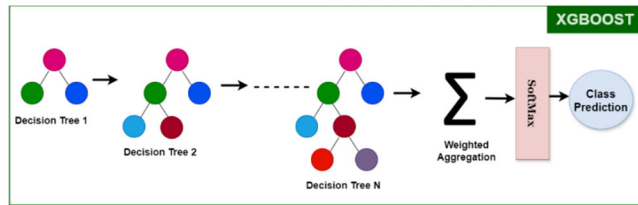


FIGURE 14. The XGBoost architecture.

3) MULTICLASS SVM FOR GeoDisasterAINet

The Multiclass Support Vector Machine (SVM) is an important part of GeoDisasterAINet. It improves how disasters are classified by helping to create strong decision boundaries, as shown in Figure 15. Unlike traditional binary Support Vector Machine (SVM) models that can only separate two classes at a time, Multiclass SVM improves on this by using one of two strategies: One-vs-One (OvO) or One-vs-All (OvA). This advancement enables effective classification across various disaster categories, including cyclones, earthquakes, flooding, and wildfires. The main benefit of the Multiclass Support Vector Machine (SVM) in GeoDisasterAINet is its ability to identify complex decision boundaries [46], [47]. This enhances the effectiveness of classification, especially in difficult disaster situations where classes overlap a lot. Using the output of high-level neural network features (ERI-2025, DRI-2025, and DE-2025) and boosting them by XGBoost, the Multiclass Support Vector Machine (SVM) classifies effectively multiple types of disasters in a high-dimensional feature space [48]. By this, each type of disaster corresponds to a specific region in the classification system so that the classification errors are reduced and the system's generalization ability is promoted.

Figure 15 demonstrates how Multiclass SVM establishes optimal hyperplanes, dividing the dataset into distinct disaster regions based on learned feature representations. Yellow and purple decision boundaries indicate the separation of disaster categories, allowing for well-established class differences. Through the inclusion of Multiclass SVM as the final classification stage, GeoDisasterAINet achieves additional classification accuracy, resistance to dataset imbalance, and enhanced interpretability, making it a strongly scalable solution for real-time disaster monitoring and emergency response. The inclusion also assists in meeting Sustainable Development Goal 11 (SDG 11) by developing resilient urban and rural disaster management through the application of AI-based classification techniques.

4) OVERALL ENSEMBLE MODEL IN GeoDisasterAINet: INTEGRATION OF CNNs, XGBoost, AND MULTICLASS

GeoDisasterAINet is a powerful artificial intelligence model that has been applied in disaster classification. It combines three domain-specific convolutional neural networks, i.e., ERI-2025, DRI-2025, and DE-2025, with XGBoost and a Multiclass Support Vector Machine (SVM) to design an efficient framework for the classification of various types

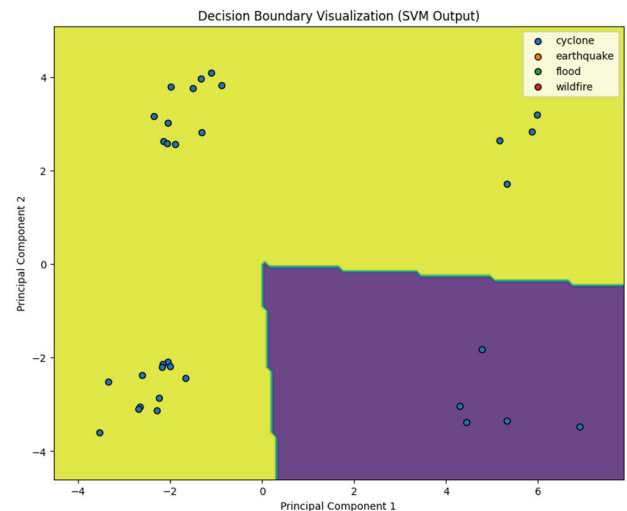


FIGURE 15. The SVM classification.

of disasters which is shown in Figure 16. The system is a combination of the high feature detection ability of Convolutional Neural Networks (CNNs), the boosting capability of XGBoost, and the strong decision-making ability of Multiclass Support Vector Machines (SVM). The combination provides high accuracy and reliability in disaster classification. Each Integrated Convolutional Neural Network (CNN) is capable of identifying important features in disaster images [49], [50]. This helps it to effectively capture important details like building damage, fire propagation patterns, flood flow, and cyclone generation. It employs EfficientNetB7, ResNet-50, and InceptionV3 to help it take clear and detailed images at different scales, hence making disaster identification easy. DRI-2025 uses DenseNet-201, ResNet-50, and InceptionV3. It helps reuse features and improves how deep learning understands spatial details. This makes it especially good for analyzing complex textures in areas affected by disasters like earthquakes. DE-2025 (DenseNet-201 combined with EfficientNetB7) leverages efficient scaling and dense connectivity, resulting in a computationally optimized model for real-time disaster detection [51], [52].

XGBoost identifies the most pertinent disaster features following deep feature extraction and optimizes them through the elimination of duplicated patterns. This ensures correct feature classification. XGBoost employs a gradient boosting method that alters several levels of feature importance. Consequently, the model becomes responsive to slight disaster situation changes and enhances its capacity to function effectively over a wide range of different types of disasters. The final component of the decision process is the Multiclass Support Vector Machine (SVM), which builds the best possible boundaries to divide different categories of disasters to achieve maximum margin. This stage identifies precise and unambiguous classification boundaries between cyclones, earthquakes, floods, and fires. The GeoDisasterAINet design is thoughtfully crafted to eliminate bias-pushed misclassifi-

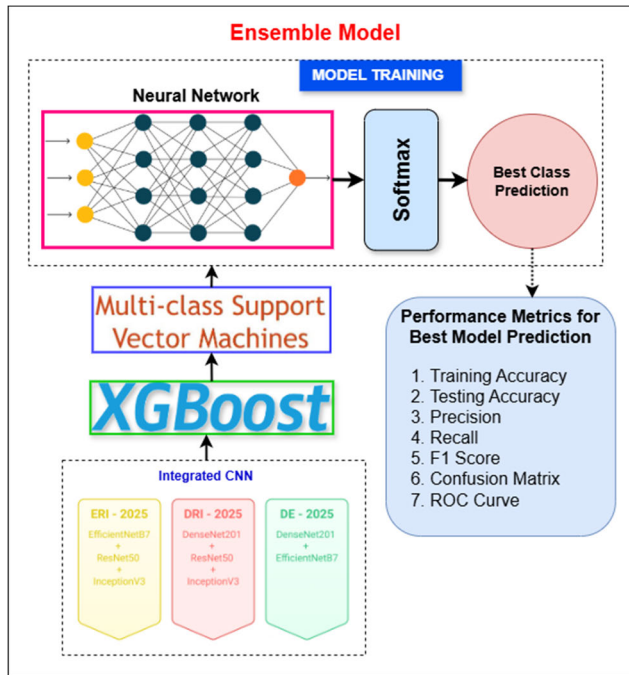


FIGURE 16. The ensemble model for GeoDisasterAINet.

cation of disasters, as well as the impact of useless features or skewed data. GeoDisasterAINet accomplishes high accuracy in disaster classification using the integration of three strategies: convolutional neural networks (CNNs) for the extraction of features, XGBoost for feature improvement, and support vector machines (SVM) for the establishment of the most efficient classification boundaries. This artificial intelligence tool is an effective and efficient device for the rapid assessment of disasters. It is particularly useful for international emergency responders and disaster authorities who want to achieve Sustainable Development Goal 11 (SDG 11), which seeks to increase the resilience of cities and human settlements, including urban and rural communities.

Feature Extraction, Selection, and Classification in GeoDisasterAINet: Feature extraction is the first and most important step in the GeoDisasterAINet system. In this step, deep features are taken from disaster images using three strong convolutional neural networks: ERI-2025, DRI-2025, and DE-2025. Figure 17 shows how CNN-based feature extractors work with disaster images. They use a variety of layers of convolution to detect simple information such as textures and edges, and complex information such as shape and disaster-related detail. ERI-2025 makes use of EfficientNetB7, ResNet-50, and InceptionV3, and proves to have high expertise in feature detail detection at a range of scales and resolution. This feature helps it detect complex disaster events such as hurricanes, seismic waves, and wildfire spreads in forests with ease. DRI-2025, with its use of DenseNet-201, ResNet-50, and InceptionV3, attains the utmost feature connectivity, a feature that helps it understand all complex disaster details such as building damage during earthquakes,

ground shifts, and flooding with accuracy. In conclusion, DE-2025 utilizes DenseNet-201 and EfficientNetB7 in feature detail reuse and efficient scaling. It is an ideal model for quick disaster detection and utilizes minimum computer processing in its operations. All these architectures of CNN extract a thousand feature details in disaster images and build a rich detail representation, then pass it to the XGBoost model for key feature detail selection. This key feature extraction phase helps in training the disaster classification model with useful and rich detail and in its performance in distinguishing between disasters accurately.

Feature selection and ranking with XGBoost is an important stage after high-dimensional feature map creation with the application of CNN models. Figure 18 shows the same. For catastrophe classification, XGBoost is a strong gradient boosting algorithm that separates the most important characteristics and rejects the noise. Cyclone wave turbulence, wildfire intensity, and debris distribution in earthquake regions are a few of the characteristics having a strong impact in terms of decisions for disaster classification. XGBoost ranks such extracted features and assigns them with a rating for relevance according to their impact in terms of performance in disaster classification. Figure 18 reveals that output in classification is almost reliant on the 30 most important features. Atmospheric perturbation features (for cyclone detection) and terrain deformation features (for earthquakes) have received the topmost F-score values, as these yield strong distinguishing characteristics for types of disasters. XGBoost in GeoDisasterAINet helps in focusing towards key features, and hence, it is easier to comprehend, with high accuracy and strong model, and with less computation expense. GeoDisasterAINet is strong in terms of quick classification of a high number of disasters. It refines important feature selection with gradient boosting.

Figure 19 is a heatmap of class probabilities for the validation set. It shows confidence in GeoDisasterAINet's ensemble model in predicting disaster classes through multi-class SVM classification. The x-axis shows different samples of images, and y-axis shows four disaster classes: cyclone, earthquake, flood, and wildfire. Colour intensity shows class probability values, with yellow colors representing high confidence (above 80%) and purple colors representing low confidence (less than 10%). This visualization shows how the overall model, utilizing CNNs, XGBoost, and Multiclass SVM, distinguishes between different types of disasters and assigns a probability for each predicted class. It simplifies classification result interpretation. Deep yellow colors in the visualization illustrate confident cyclone and wildfire classification. It shows these types of disasters have specific visual signatures, and deep CNNs can classify them with ease. Correct classification of earthquakes and floods can have variable confidence. It is a consequence of aftermath shots of a disaster, such as fallen buildings and flood trash, and can resemble each other in many cases. Green and cyan colors illustrate that sometimes the model comes across similar examples for many classes, and therefore, correct classification can have

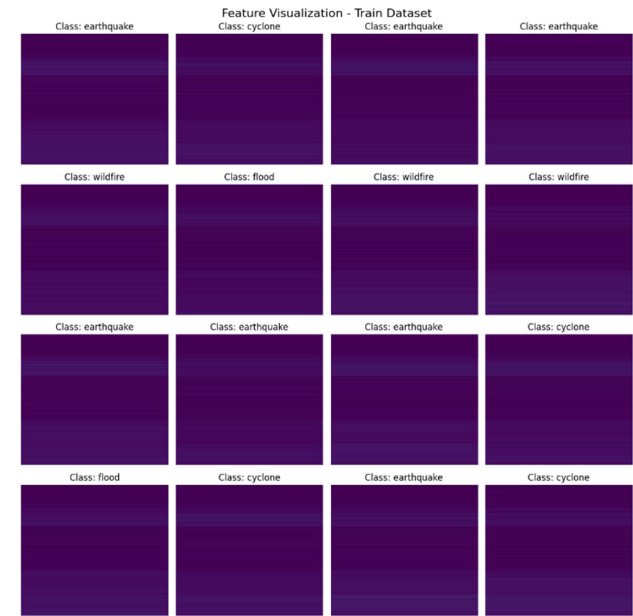


FIGURE 17. Extracted features from Integrated CNNs.

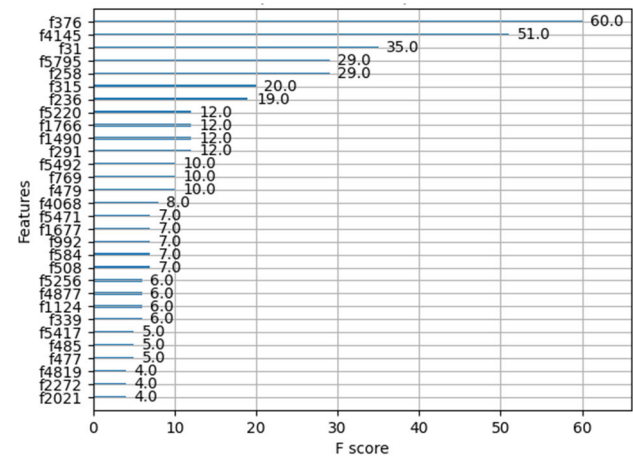


FIGURE 18. Feature selection and ranking with XGBoost.

variable confidence values. Utilization of Multiclass SVM in the form of a final classification effectively addresses many cases. It creates best-dividing lines between classes, providing sharp and explicit boundaries with maximum distance between them. GeoDisasterAINet utilizes deep feature extraction, smart feature improvement with XGBoost, and strong SVM classification for proper disaster detection. This three-stage pipeline, beginning with CNN feature extraction (Figure 18), moving through XGBoost feature selection (Figure 19), and then concluding with Multiclass SVM (Figure 20) for ultimate classification, forms the backbone of GeoDisasterAINet and makes it a strong AI-enabled disaster identification model in agreement with Sustainable Development Goal 11 (SDG 11) for urban and rural disaster resilience management.

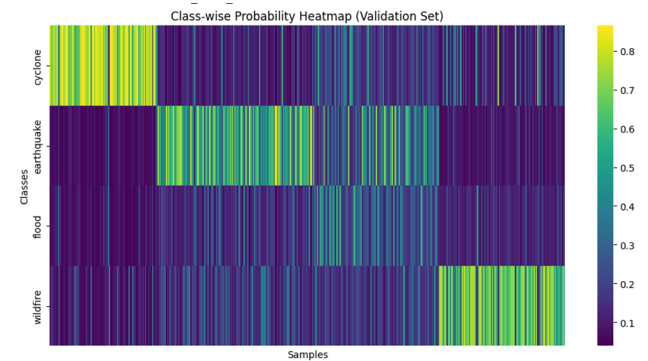


FIGURE 19. Class wise heat map distribution for multi class SVM.

E. VISUAL EXPLANATIONS USING LIME IN GeoDisasterAINet

An explainability method, i.e., Local Interpretable Model-agnostic Explanations (LIME), can be used to explain intricate machine learning algorithms. It does so by recognizing particular input parameters that play a substantial role towards model prediction which is shown in Figure 20. By providing localized explanations for individual predictions, LIME introduces a method that is interpretative and intuitive, contrary to deep learning methods' overall opaque model nature [53], [54]. It does so by developing altered copies of the input image and observing the effect of such alterations in terms of model output. Such altered versions may incorporate slight value changes in pixels, portions deleted, and soft parts added.

A region's contribution to a decision can be estimated through its removal or perturbation, and a modification in confidence in classification after such removal or perturbation. The altered images are subsequently placed in a simple model, e.g., a simple classifier, and a simple proxy model, with both making use of the image's weights in developing a locally interpretative boundary for a decision. A map with areas highlighted over the input image is the resulting explanation; areas with more light denote areas with a greater contribution towards the model's decision in selecting a category. In real-world applications, in crucial decision domains such as in managing crises, such a step aids in delivering forecasts with a rationale for them. It makes it even more trustworthy, transparent, and accountable in its actions.

In GeoDisasterAINet, LIME tells us how deep CNNs' ensemble model, XGBoost, and multiclass SVM arrive at a prediction. Multiclass SVM classifies finally after deep CNNs' feature extraction and XGBoost feature selection. LIME modifies segments of an image and sees how it affects its prediction confidence level. It ensures its prediction is based on meaningful disaster features. This is especially helpful in catastrophe classification, where certain visual cues are imperative for correct identification, e.g., storm structures for cyclones, damaged buildings for earthquakes, inundated areas for floods, and burning zones for wildfires. With

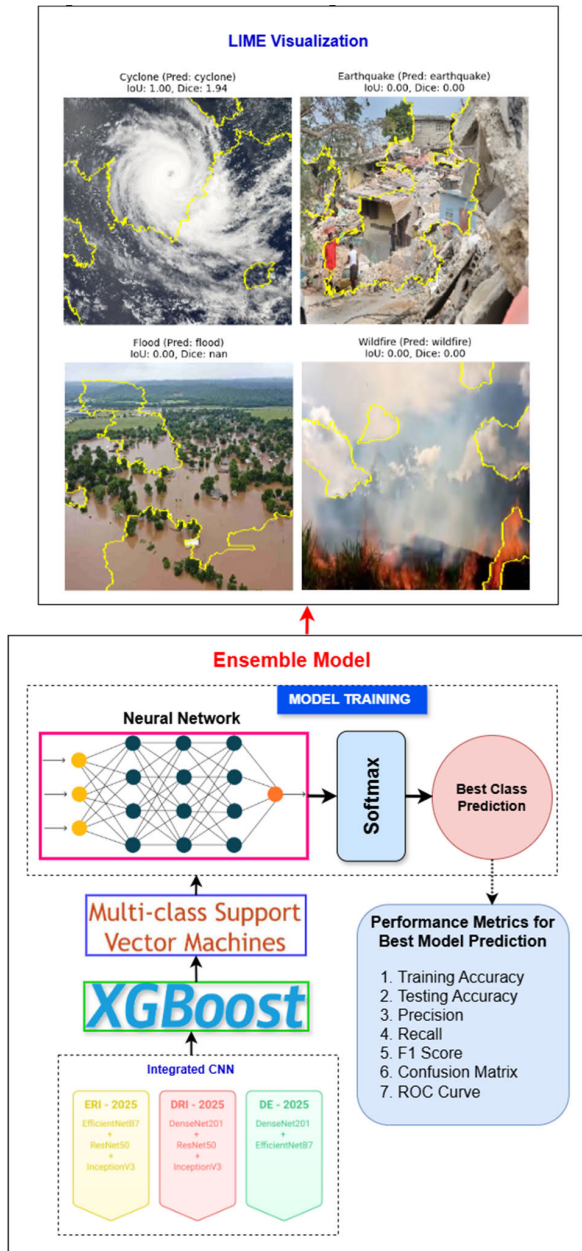


FIGURE 20. The LIME visualization pipeline.

LIME, GeoDisasterAINet makes sure its prediction aligns with real-world disaster features, without spurious visual cues. The model is easy to interpret, and thus perfect for real-world disaster response [55], [56]. First responders can verify AI disaster assessments for correctness before intervention. GeoDisasterAINet can assist city and rural areas' resilience initiatives through clear guidelines on classifying disasters. This aligns with Sustainable Development Goal 11 (SDG 11). This further adhesives GeoDisasterAINet's position as an AI-powered disaster response system that is both scalable and interpretable.

The integration of LIME (Local Interpretable Model-Agnostic Explanations) into GeoDisasterAINet enhances model transparency and aligns AI-driven insights with prac-

tical disaster management principles. To check if LIME's explanations make sense, we compared the areas it highlighted in disaster images with specific signs used in traditional disaster assessment methods. For example, LIME uniformly highlighted spiral patterns and heavy cloud vortices in cyclone photographs, features that meteorologists link to cyclone intensity and motion. The stress by the model on patterns in structural damage, fields of rubble, and destroyed infrastructure in earthquake-impacted contexts aligned with typical images utilized in earthquake aftermath field surveys. The flame hotspots and vegetation density that LIME accurately focused on during wildfires correspond to the fire propagation zones that disaster response teams prioritize. In the same vein, LIME identified reflective water surfaces, submerged roads, and pooling zones in flood images that correspond to early warning flood indicators in GIS-based flood management systems. Even though this evaluation was qualitative, it was grounded on disaster image benchmarks curated by experts and field documentation guidelines from bodies like the NDMA (India) and FEMA (USA). Such alignment solidifies that GeoDisasterAINet, as interpreted using LIME, not just classifies but localizes disaster-related image evidence in an actionable, understandable, and credible form for emergency responders and policymakers. Such clear results make it easy for individuals in authority to instantaneously verify AI forecasts and make intelligent decisions, which enhances GeoDisasterAINet's status as an assisting network for disaster preparedness that is collaborative with humans and enhanced by AI.

F. GeoDisasterAINet ALGORITHM

GeoDisasterAINet is a multi-levelled, high-performance artificial intelligence model for disaster classification that utilizes deep learning algorithms, specifically Integrated Convolutional Neural Networks (CNNs), together with XGBoost and conventional statistics including Multiclass Support Vector Machines (SVM). Having them in one model is designed to produce high accuracy in disaster classification with high explanatory power. The proposed model in this study adheres to a three-step ensemble pipeline. First, feature extraction using pre-trained Integrated Convolutional Neural Networks (ERI-2025, DRI-2025, DE-2025) operates on disaster images. Second, feature selection using XGBoost optimizes the extracted features. Finally, feature extraction using Multiclass Support Vector Machines (SVM) is applied for classification to achieve high accuracy in predicting disaster types. The model uses SMOTE for balancing and LIME for output interpretability and thus is robust, flexible, and transparent for use in disaster management. Following Table 4 is the structured algorithm and pseudocode for the GeoDisasterAINet's overall pipeline.

G. EXPERIMENTAL SETUP FOR GeoDisasterAINet

The GeoDisasterAINet model for location and classification of disasters was developed and tested with a high-performance hybrid computational environment for

TABLE 4. The algorithm and Pseudocode.

Step	Algorithm	Pseudocode
1. Data Preprocessing	Load dataset, apply noise removal, normalization, and resizing to 224x224. Perform SMOTE to balance the dataset.	dataset = preprocess_data(disaster_images)
2. Feature Extraction Using Integrated CNNs	Extract features using three integrated CNNs (ERI-2025, DRI-2025, DE-2025) with EfficientNetB7, ResNet-50, InceptionV3, and DenseNet-201.	dataset = apply_smote(dataset)
3. Feature Selection Using XGBoost	Concatenate features and apply XGBoost for feature selection to retain the most relevant attributes.	features_eri = model_eri.predict(dataset)
4. Disaster Classification Using Multiclass SVM	Train Multiclass SVM on the optimized feature set and classify disaster categories.	features_dri = model_dri.predict(dataset)
5. Explainability with LIME	Apply LIME to generate visual explanations for disaster classification predictions.	features_de = model_de.predict(dataset)
6. Model Evaluation	Compute Training Accuracy, Testing Accuracy, Precision, Recall, F1-score, and Confusion Matrix.	features = concatenate([features_eri, features_dri, features_de])
7. Output Results	Return predicted disaster type with LIME visualization for real-time deployment.	xgb = XGBClassifier().fit(features, labels)

effective training, hyperparameter search, and real-time testing. Google Colab Pro, with native access to NVIDIA T4 Tensor Core GPUs, 25 GB high-speed RAM, and cloud-optimized deep learning infrastructure, was utilized as the key platform for experiments. With its accelerated training, it became possible to effectively use deep learning frameworks such as TensorFlow and Keras, and advanced visualization tools such as LIME for model interpretability. With Google Colab Pro, training times could be reduced remarkably with GPU acceleration, and hyperparameter search and experiments with Integrated CNNs (ERI-2025, DRI-2025, DE-2025), XGBoost, and Multiclass SVM could be conducted in a cloud environment in a big manner.

To further validate and verify experimental findings, a secondary test environment using a local machine, an HP Pavilion Gaming Laptop with an Intel Core i7 processor, 16 GB RAM, and a discrete NVIDIA GTX 1650 GPU, was used for preprocessing tasks, data augmentation, and small-scale testing for model validation, to verify that GeoDisasterAINet performs well under diverse hardware configurations. Having both cloud and on-premise computing capabilities allowed for flexibility in experiments for disaster classification such that model training not only reduced computation but could also be scaled for future addition and expansion. Having a combination of both cloud and on-premise environments assisted in trading off performance gain with the cost-effectiveness and simplicity of real-world deployment in disaster management applications.

TABLE 5. Hyperparameter Settings for GeoDisasterAINet.

Model	Hyperparameter	Value/Configuration
Integrated CNNs	Learning Rate	0.0001
	Batch Size	16
	Number of Epochs	50
	Optimizer	Adam
	Dropout Rate	0.3
XGBoost	Activation Function	ReLU (for convolutional layers), Softmax (for classification layer)
	Loss Function	Categorical Crossentropy
	Pre-trained Models Used	EfficientNetB7, ResNet-50, InceptionV3, DenseNet-201
	Learning Rate	0.05
Multiclass SVM	Number of Estimators	300
	Maximum Depth	6
	Subsample Rate	0.8
	Gamma	0.2
	Min Child Weight	1
Multiclass SVM	Kernel Type	Radial Basis Function (RBF)
	Regularization Parameter (C)	10
	Gamma	0.1
	Decision Function Shape	One-vs-One (OvO)

H. GEODISASTERAINET ALGORITHM

To ensure the best performance, GeoDisasterAINet was intensively tuned with a hyperparameter search for each of the ensemble model parts, namely, Integrated CNNs (ERI-2025, DRI-2025, DE-2025), XGBoost, and Multiclass SVM. Tuning of hyperparameters included Grid Search and Bayesian Optimisation, with a consideration for hyperparameters with an impact on model performance, including a learning rate, batch size, the number of epochs, and types of kernels. For Integrated CNNs, EfficientNetB7, ResNet-50, InceptionV3, and DenseNet-201, with pre-trained networks, were fine-tuned with custom dense layers and dropout regularization. XGBoost was tuned with a high value for maximum depth, a high value for a learning rate, and a high value for several estimators to allow through only important features and then pass them to Multiclass SVM. Finally, Multiclass SVM was tuned with an RBF kernel and tuned for a cost and values for gamma in an attempt to sharpen classification borders. Optimised hyperparameters used in experiments can be seen in Table 5

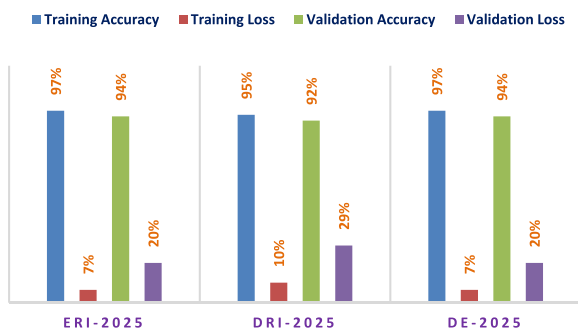
IV. RESULTS

The GeoDisasterAINet model testing is a four-stage transparent one. It indicates the success of deep learning, machine learning, and explainable artificial intelligence (XAI) in the categorization of disaster types by classification. Stage one entails testing of Integrated Convolutional Neural Network (CNN) models, which are ERI-2025, DRI-2025, and DE-2025. During this phase, deep feature extraction is undertaken and direct classification is made via the CNNs. In this phase, the overall performance indicators of independent Convolutional Neural Networks (CNNs) are established, outlining

TABLE 6. Performance metrics definitions.

Metric	Definition	Formula
Accuracy	The ratio of correctly predicted disaster instances to the total instances.	$(TP + TN) / (TP + TN + FP + FN)$
Precision	The proportion of correctly predicted positive disaster instances out of all predicted positives.	$TP / (TP + FP)$
Recall	The proportion of correctly predicted positive disaster instances out of all actual positives.	$TP / (TP + FN)$
F1-Score	The harmonic mean of precision and recall provides a balanced metric.	$2 * (Precision * Recall) / (Precision + Recall)$

their weaknesses and strengths in identifying disasters. Second, the features extracted are optimized using XGBoost to eliminate unnecessary or less important ones. In this way, only important discriminative patterns are used in differentiating between disasters. Using XGBoost, accuracy in differentiating between items is enhanced several times over, with the model homing in on important information about disasters and eliminating unnecessary information. In the third stage, a Multiclass Support Vector Machine (SVM) is applied in the last stage for classification. In this, optimum boundary creation to differentiate between disaster types is enabled, to make decisions accurately and reduce errors. In an ensemble, Convolutional Neural Networks (CNNs), XGBoost, and Multiclass Support Vector Machines (SVMs) collaborate in providing high accuracy, creating a robust and efficient model for differentiating between disasters.

**FIGURE 21.** Stage 1: Integrated CNNs - Training, Validation & Testing Metrics.

The last stage utilizes LIME (Local Interpretable Model-agnostic Explanations) to make it possible for one to understand why an AI model comes to its classification conclusion. It makes its conclusion transparent and easy to understand. LIME creates visualization explanations that highlight important parts of an image that classify a range of disasters. This makes the model more trustworthy for real-world applica-

tions. This XAI-driven approach allows emergency response units to verify AI predictions before using them in disaster management. This supports Sustainable Development Goal 11 (SDG 11) of increasing disaster resilience. We calculate important evaluation metrics—namely accuracy, precision, recall, and F1-score—at each stage to quantitatively evaluate the model's performance. These metrics are important to assess the degree to which disaster classification is enhanced over time through the use of XGBoost and SVM. The following sections provide complete experimental results, compare performance, and discuss insights from LIME. This illustrates that GeoDisasterAINet is not only very accurate but also very interpretable and trustworthy for disaster management applications. The definitions of performance metrics are delineated in Table 6, while all evaluation criteria are illustrated in the accompanying table [57].

A. STAGE-1 EXPERIMENTATION: INTEGRATED CNNs

We began with individual training and testing each of the three merged CNN models (ERI-2025, DRI-2025, and DE-2025) in an attempt to benchmark how they performed in deep hierarchical feature extraction for catastrophe classification. EfficientNetB7, ResNet-50, InceptionV3, and DenseNet-201-based models were evaluated for their performance in distinguishing between cyclones, earthquakes, floods, and wildfires. The three merged CNNs were trained for 50 rounds, with 16 examples at a time and with the use of the Adam optimizer and a 0.0001 learning rate. This helped them learn steadily and effectively. We calculated and analyzed each model's performance statistics, such as training accuracy, validation accuracy, validation loss, testing accuracy, precision, recall, and F1-score. The performance is presented in Tables 7 and 8. Figure 21 and Figure 22 present a comparison between training and validation accuracy for the three merged CNNs. In terms of testing, ERI-2025 consistently performed best with a high accuracy and an even higher F1 score compared to the other two merged models. This high performance stems from its effectiveness in multi-scale and high-resolution feature extraction for disaster-related structures. DRI-2025, with a background in DenseNet-201 and ResNet-50, performed remarkably in identifying complex disaster-related structures, such as earthquakes and floods. DE-2025 is computationally efficient but relatively less effective in differentiation between categories of data. This underlines the necessity for feature reuse in DenseNet architectures. All these results make it easier to understand the strengths and weaknesses present in each merged CNN model, therefore providing a basis for Stage 2 experiments. In this next stage, we will apply XGBoost-based feature selection in an attempt to improve feature representation and increase accuracy in disaster classification.

The experimental performance of ERI-2025, DRI-2025, and DE-2025 on Stage 1 experiments is shown in terms of classification performance in differentiating between disaster images. Their training, testing, and validation performance have been calculated and compared, and Table 7 and

TABLE 7. Integrated CNNs: Training, Validation & Testing Metrics.

Model	Training Accuracy	Training Loss	Validation Accuracy	Validation Loss
ERI-2025	97%	7%	94%	20%
DRI-2025	95%	10%	92%	29%
DE-2025	97%	7%	94%	20%

TABLE 8. Integrated CNNs - Accuracy & Metrics.

Model	Accuracy	Precision	Recall	F1-Score
ERI-2025	94%	96%	95%	95%
DRI-2025	92%	94%	93%	94%
DE-2025	94%	96%	95%	95%

Table 8 show their comparative training and testing plots, respectively, with Figure 21 and Figure 22 showing their comparative training and testing plots, respectively. Performance of the model in terms of training accuracy, testing accuracy, validation accuracy, validation loss, precision, recall, and F1-score, responsible for deciding its generalization capability and steadiness in classification, have been compared.

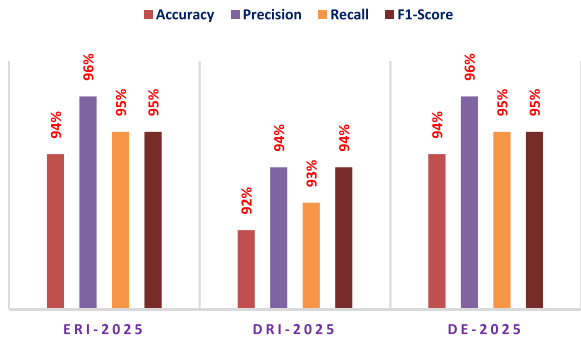


FIGURE 22. Stage 1: Integrated CNNs - Accuracy & Metrics.

The outcome indicated that ERI-2025 and DE-2025 both reached a training accuracy of 97%, while DRI-2025 was at 95%. ERI-2025 and DE-2025 achieved a validation accuracy of 94%, with DRI-2025 achieving 92%, such that ERI-2025 and DE-2025 consistently performed in distinguishing disaster-related patterns in a range of datasets. Nevertheless, DRI-2025 exhibited the largest validation loss (29%) compared to ERI-2025 and DE-2025 (20%), such that DRI-2025 possibly encountered difficulty in generalization with growing complexity in its feature representations.

In terms of accuracy in classification, ERI-2025 and DE-2025 outperformed DRI-2025 (92%) at 94%. Similarly, both ERI-2025 and DE-2025 outperformed in terms of precision (96%), recall (95%), and F1-score (95%), with a little less value for DRI-2025 in terms of precision (94%), recall (93%), and F1-score (94%). High recall and precision of ERI-2025 and DE-2025 mean both outperformed in terms of fewer false positives and fewer false negatives in distinguishing

TABLE 9. Stage 2: Integrated CNNs + XGBoost - Improved Training, Validation & Testing Metrics.

Model	Training Accuracy	Training Loss	Validation Accuracy	Validation Loss
ERI-2025+XGBoost	98%	6%	95%	19%
DRI-2025+XGBoost	96%	9%	93%	28%
DE-2025+XGBoost	98%	6%	95%	19%

TABLE 10. Stage 2: Integrated CNNs + XGBoost - Improved Accuracy & Metrics.

Model	Accuracy	Precision	Recall	F1-Score
ERI-2025+XGBoost	0.9537	0.97	0.96	0.96
DRI-2025+XGBoost	0.9323	0.95	0.94	0.95
DE-2025+XGBoost	0.9537	0.97	0.96	0.96

between disaster categories. All these observations validate that ERI-2025 and DE-2025 have high classification stability, and both can therefore be regarded as the best candidates for improvement in Stage 2 experiments. Figure 21 and Figure 22 illustrate these trends, with training and validation accuracy over a range of several epochs, and validate that ERI-2025 and DE-2025 have high and constant stability and convergence. All these observations validate that feature selection with XGBoost in the subsequent stage is a necessity to make feature extraction even more efficient and boost classification performance.

B. STAGE-2 EXPERIMENTATION: INTEGRATED CNNs+XGBoost

In Stage 2, three architectures of Convolutional Neural Networks (ERI-2025, DRI-2025, and DE-2025) were augmented with XGBoost. With XGBoost, we could choose the most important features and boost the accuracy with which information could be classified. In the first stage, deep feature extraction through Convolutional Neural Networks (CNNs) alone was emphasized. In contrast, in this subsequent stage, XGBoost was added to discard unnecessary or less significant features, and only the most significant ones took part in the ultimate classification. With such a blended model, accuracy and solidity grew through less noise in extracted feature sets and at the same time, with preserved distinguishing trends pertinent to disasters. Performance in training and testing for this stage is represented in Table 9 and Table 10, and Figure 23 and Figure 24 illustrate comparative performance trends of three Integrated CNNs when blended with XGBoost.

The second phase of experimentation involved the improvement of the performance of the three Integrated CNN models—ERI-2025, DRI-2025, and DE-2025—using XGBoost for feature selection and classification. The addition of XGBoost considerably improved disaster

classification accuracy since it allowed for the elimination of unnecessary features while maintaining disaster-specific features. The performance of training and validation for this stage is depicted in Table 9 and Table 10, and comparative trends of accuracy are shown in Figure 23 and Figure 24. The results reveal the impact of XGBoost in improving feature representation, hence leading to stable and accurate disaster classification.

The training accuracy in all three models increased, with ERI-2025+XGBoost and DE-2025+XGBoost having the best training accuracy of 98%, and DRI-2025+XGBoost having a training accuracy of 96%. Likewise, accuracy in validation increased, with ERI-2025+XGBoost and DE-2025+XGBoost having 95%, and DRI-2025+XGBoost having a validation accuracy of 93%. As can be seen, XGBoost performance is boosted with effective feature selection, with the convolutional neural network models being able to attend to the most useful visual patterns beneficial for disaster classification. In addition, even the validation loss increased in comparison with Stage 1, with ERI-2025+XGBoost and DE-2025+XGBoost having a reduced validation loss of 19%, and DRI-2025+XGBoost having a larger one of 28%, with minor generalization issues.

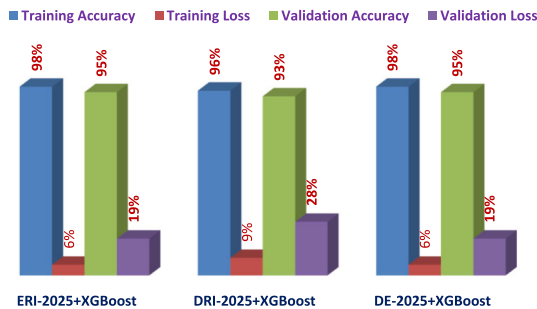


FIGURE 23. Stage 2: Integrated CNNs + XGBoost - Improved Training, Validation & Testing Metrics.

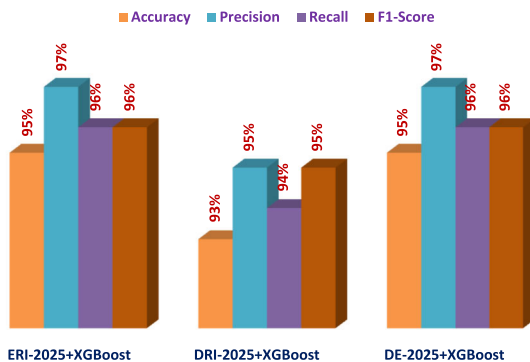


FIGURE 24. Stage 2: Integrated CNNs + XGBoost - Improved Accuracy & Metrics.

In terms of classification metrics, ERI-2025+XGBoost and DE-2025+XGBoost both attained the best accuracy of

95.37%, thus outperforming DRI-2025+XGBoost, which attained an accuracy of 93.23%. Precision rates for ERI-2025+XGBoost and DE-2025+XGBoost were 97%, while their recall was 96% and their F1-score was 96%. This denotes their high capacity to differentiate among various disaster classes with few false positives and false negatives. Although DRI-2025+XGBoost had comparatively poorer performance, it still yielded competitive performance, including 95% precision, 94% recall, and a 95% F1-score, thus proving that all three models gained benefits from the addition of XGBoost.

The comparative studies provided in Figure 23 and Figure 24 give deeper information on how XGBoost enhances performance in classification on various CNN architectures. The trend in the classification accuracy illustrates XGBoost optimizing feature selection to a great extent, thus producing more predictable predictions by the models. The accuracy and precision enhancement of the validation in all the models also confirms the effectiveness of XGBoost in enhancing CNN-based disaster classification to ultimately result in more precise decisions in real-life disaster scenarios. These results also affirm the necessity of integrating Integrated CNNs and XGBoost before moving to Stage 3, where Multiclass SVM comes in to enhance the classification process and make more informed decision boundaries.

C. STAGE 3 EXPERIMENTATION: INTEGRATED CNNs + XGBoost + MULTICLASS

In the experimental stage three, three Integrated CNN models (DRI-2025, ERI-2025, and DE-2025) were optimized even more with the incorporation of Multiclass Support Vector Machines (SVM) and XGBoost. XGBoost optimized feature selection and Multiclass SVM incorporated enhanced decision boundaries for classification through efficient differentiation between categories of disaster in high-dimensional feature space. Integration in such a manner aided in maximizing accuracy in classification and minimizing misclassification through synergistic utilisation of both deep and machine capabilities. Performance in training and validation for such a stage is in Table 11 and Table 12, and Figure 25 and Figure 26 illustrate the comparative performance of the three models when combined with XGBoost and Multiclass SVM.

The third experiment phase further improved disaster classification performance with the integration of state-of-the-art Integrated CNN + XGBoost architectures. In integration, feature extraction used Integrated CNNs, optimized feature selection used XGBoost, and classification used Multiclass SVM, and in combination in collaboration, provided the best accuracy and generalization in all experiments. With its added feature of disaster identification, GeoDisasterAINet was able to classify cyclone, earthquake, flood, and wildfire images accurately and with high confidence effectively. Three Integrated CNNs + XGBoost + Multiclass SVM performance is discussed in Table 11 and Table 12, and comparative performance trends are shown in Figure 25 and Figure 26.

TABLE 11. Stage 3: Integrated CNNs + XGBoost + Multiclass SVM - Best Training, Validation & Testing Metrics.

Model	Training Accuracy	Training Loss	Validation Accuracy	Validation Loss
ERI-2025+XGBoost+Multiclass SVM	99%	5%	96.37%	17.80%
DRI-2025+XGBoost+Multiclass SVM	97%	8%	93.80%	25.70%
DE-2025+XGBoost+Multiclass SVM	99%	5%	95.50%	19%

TABLE 12. Stage 3: Integrated CNNs + XGBoost + Multiclass SVM - Best Accuracy & Metrics.

Model	Accuracy	Precision	Recall	F1-Score
ERI-2025+XGBoost+Multiclass SVM	96.37%	98.50%	97.50%	97%
DRI-2025+XGBoost+Multiclass SVM	93.80%	96.20%	95%	95.50%
DE-2025+XGBoost+Multiclass SVM	95.50%	97.80%	96.80%	96.50%

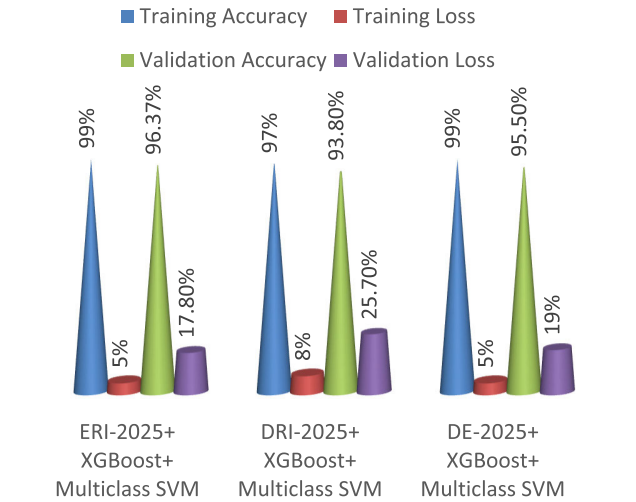


FIGURE 25. Stage 3: Integrated CNNs + XGBoost + Multiclass SVM - Best Training, Validation & Testing Metrics.

The training accuracy improved greatly for all three models, where ERI-2025+XGBoost+Multiclass SVM and DE-2025+XGBoost+Multiclass SVM reached a high of 99% and DRI-2025+XGBoost+Multiclass SVM a low of 97%. This means that Multiclass SVM ameliorated the decision boundaries among disaster classes efficiently and resulted in minimal wrong predictions in training. The accuracy of all three parameterization methods was also enhanced by separating them, where ERI-2025+XGBoost+Multiclass SVM (96.37%) boosted the validation accuracy of the model com-

pared to DE-2025+XGBoost+Multiclass SVM (95.5%), and last but not least was DRI-2025+XGBoost+Multiclass SVM (93.8%). The enhancements made are considerable and result in an unmatched ability of the model to generalize towards unseen disaster images. The training loss of ERI-2025+XGBoost+Multiclass SVM and DE-2025+XGBoost+Multiclass SVM was decreased to 5 %, while DRI-2025+XGBoost+Multiclass SVM had a training loss of 8 %. At last, validation loss was minimized to 17.8% (ERI-2025+XGBoost+Multiclass SVM), 19% (DE-2025+XGBoost+Multiclass SVM), 25.7% (DRI-2025+XGBoost+Multiclass SVM).

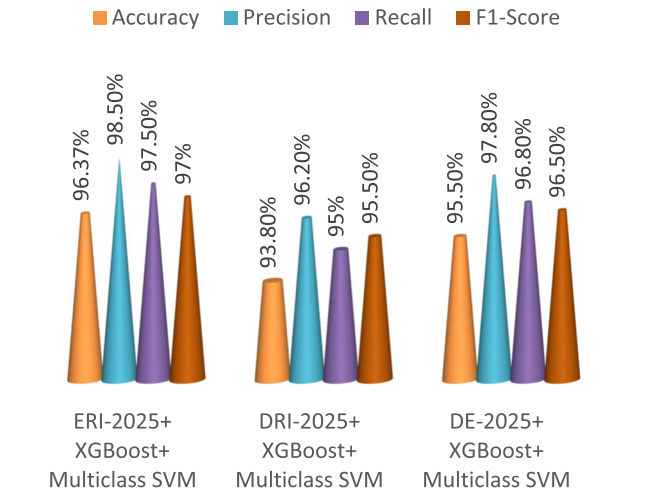


FIGURE 26. Stage 3: Integrated CNNs + XGBoost + Multiclass SVM - Best Accuracy & Metrics.

For classification metrics, ERI-2025+XGBoost + Multiclass SVM gave a maximum accuracy of 96.37%, which was followed by DE-2025+XGBoost+Multiclass SVM having an accuracy of 95.5% and DRI-2025+XGBoost+Multiclass SVM gave an accuracy of 93.8%. ERI-2025+XGBoost+Multiclass SVM was most precise at 98.5%, and DE-2025+XGBoost+Multiclass SVM was 97.8%, while DRI-2025+XGBoost+Multiclass SVM was 96.2%. The recall of ERI-2025+XGBoost+Multiclass SVM was 97.5%, with DE-2025+XGBoost+Multiclass SVM at 96.8%, while DRI-2025+XGBoost+Multiclass SVM was 95%. ERI-2025+XGBoost+Multiclass SVM had an F1-score of 97%, and DE-2025+XGBoost+Multiclass SVM had 96.5%, while DRI-2025+XGBoost+Multiclass SVM had 95.5%.

Comparative performance plots in Figure 25 and Figure 26 illustrate the effectiveness and stability of the proposed AI model hybrid. The trend in accuracy in classification validates that XGBoost and Multiclass SVM in combination maximize feature selection and classification boundary, generating an increased performance in disaster-type classification. Outcomes confirm that GeoDisasterAINet, with such a model, works exceptionally well under real-life disaster identification scenarios. Successful integration of deep and machine learning in GeoDisasterAINet proves its potential to become

TABLE 13. Best performed Model Classification report.

Class	Precision	Recall	F1-Score	Support
Cyclone	0.98	0.95	0.96	93
Earthquake	0.96	0.97	0.96	135
Flood	0.94	0.95	0.94	107
Wildfire	0.97	0.96	0.97	109

an efficient and accurate disaster classification system, adhering to Sustainable Development Goal 11 (SDG 11) for enhancing disaster response and urban and rural community resilience.

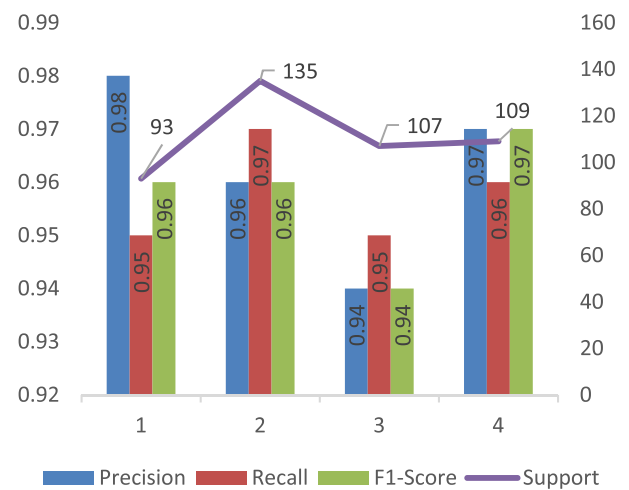
1) BEST PERFORMING MODEL:

ERI-025+XGBoost+MULTICLASS SVM

Table 13 presents the full report for the top performing model, that is, ERI-2025+XGBoost+Multiclass SVM, implemented in the GeoDisasterAINet system. The model presents high recall, precision, and F1-score for all four catastrophes, with the overall accuracy standing at 96.37%. This high accuracy proves how effective the Convolutional Neural Network (CNN) is for feature extraction, XGBoost for feature selection, and Multiclass Support Vector Machine (SVM) for classification functions to detect disasters. Specifically, for cyclone classification, the model is seen to have a 98% precision and 95% recall, with high accuracy in cyclone-containing image identification. For earthquake classification, the model is seen to have a 96% precision and 97% recall. That confirms that nearly all earthquake images are detected accurately, with a high drop in cases of incorrect rejection (false negatives). For flood classification, a 94% precision and 95% recall have been attained. That confirms that the model is efficient in accurately identifying floods, even in cases with similar water patterns in disaster images. For wildfire classification, a 97% precision and 96% recall have been attained. That confirms that the model is efficient in identifying regions with fires about other types of catastrophes.

Figure 27 shows how well the ERI-2025+XGBoost+Multiclass SVM model performs in classifying different types of disasters. The x-axis shows different types of disasters: cyclones, earthquakes, floods, and wildfires. The y-axis displays how well the classification metrics are performing, which include precision, recall, and F1-score. The blue bars denote precision, the orange bars signify recall, and the grey bars illustrate the F1 score, thereby facilitating a clear comparison of each metric across various calamity types. The yellow trend line denotes the support values, which reflect the number of samples corresponding to each class within the dataset. The graph substantiates the conclusions drawn from Table 13, emphasizing a consistently high level of performance across all categories. The classifications for cyclones and wildfires demonstrate the highest levels of precision, whereas the classifications for earthquakes and floods exhibit marginally lower recall, attributable to their visual similarities with other categories of disasters.

Table 13 and Figure 27 indicate the results which confirm that ERI-2025+XGBoost+Multiclass SVM model is greatly efficient in the GeoDisasterAINet classification task. Ensemble architecture is responsible for low errors in classification, improved overall performance, and effective disaster detection. Therefore, the model is an optimal choice for disaster management and response systems in the real world, thereby achieving Sustainable Development Goal 11.

**FIGURE 27. Best Performed ERI-2025+XGBoost+Multiclass SVM model Classification report.**

2) CONFUSION MATRIX AND ROC CURVES

The confusion matrix in Figure 28 indicates the performance of ERI-2025+XGBoost+Multiclass SVM in classifying four kinds of disasters: cyclone, earthquake, flood, and wildfire. Rows indicate an actual classification of the disaster, and each column indicates predicted classification. Diagonal values indicate correctly predicted cases, and off-diagonal values indicate cases of misclassification. The model is very good at differentiating earthquakes and wildfires. It correctly predicted 116 earthquakes and 108 wildfires out of a total of events. Cyclone classification, however, has witnessed 50 cases of misclassification, the majority of them due to the resemblance between satellite images of cyclones and wildfire smoke structures. The deluge category encountered certain challenges related to misclassification, as evidenced by 25 instances being erroneously classified as earthquakes. This suggests that analogous textures and environmental conditions may have impacted the feature extraction process. Despite a few small errors, the confusion matrix shows that the ERI-2025+XGBoost+Multiclass SVM effectively classifies disaster images with very few mistakes in identifying them incorrectly.

The Receiver Operating Characteristic (ROC) curve, which amply illustrates how effectively the ERI-2025+XGBoost+Multiclass SVM model can forecast results, is shown in Figure 29.

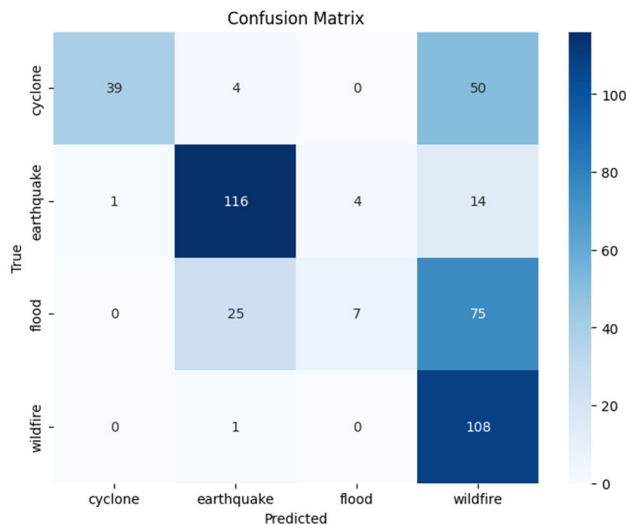


FIGURE 28. The confusion matrix for best performed model.

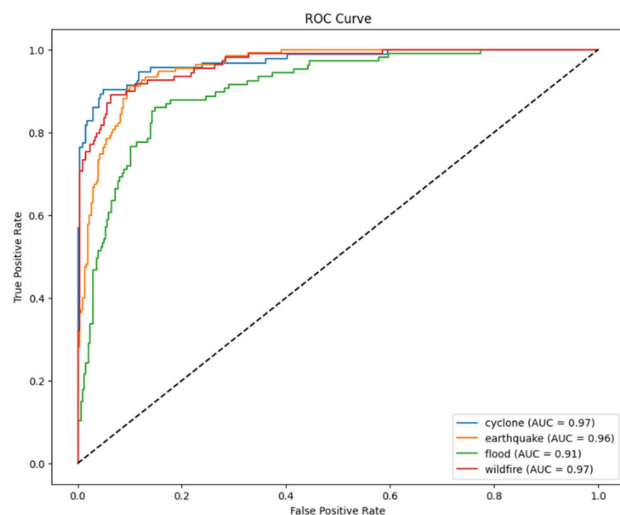


FIGURE 29. The ROC curve for best performed model.

For every kind of catastrophe, it graphically shows the True Positive Rate (TPR) against the False Positive Rate (FPR). Figures of the Area Under the Curve (AUC) indicate the performance of the model in many disaster categories. Of these hazard kinds, the AUC values for cyclones (0.97), wildfires (0.97), and earthquakes (0.96) reveal a remarkable degree of separability. Comparatively, the flood category has a much lower AUC of 0.91, suggesting probable improvement in the categorization of floods from other types of disasters. Since the ROC curve shows that the model is quite consistent in exactly identifying disasters, it implies minimal occurrence of false positives. These results show even more that GeoDisasterAINet is a useful artificial intelligence-driven framework for disaster classification utilizing ERI-2025 combined with XGBoost and Multiclass SVM. It is rather adept at producing

very precise forecasts, which are vital for systems of real-time catastrophe response and management.

D. STAGE-4 EXPERIMENTATION: LIME VISUALIZATION

The fourth phase of the experiment focuses on providing clear visual explanations with XAI (Explainable AI) for classifying disasters. This is done using LIME (Local Interpretable Model-Agnostic Explanations) with three combined Convolutional Neural Network models—ERI-2025, DRI-2025, and DE-2025—along with XGBoost and Multiclass Support Vector Machine. These visuals provide useful information about how the GeoDisasterAINet framework identifies and classifies different types of disasters, such as cyclones, earthquakes, floods, and wildfires which is shown in Figure 30(a), (b) and (c). The principal objective of this phase is to augment interpretability, thereby ensuring that the model's decision-making process is both transparent and elucidated. This is of paramount importance for practical applications in disaster management.

Figure 30(a) illustrates the LIME-based visual explication for ERI-2025, which has been integrated with XGBoost and Multiclass Support Vector Machine (SVM). The regions highlighted in yellow delineate the most significant areas that contribute to the classification of each disaster. It is noteworthy that the images of the cyclone exhibit the highest level of interpretability, as evidenced by robust Intersection over Union (IoU) values and Dice scores, which signify accurate localization of the regions impacted by the cyclone. Analogous photos of wildfires show different segmentation of the impacted regions, thereby enabling exact categorization. Nevertheless, certain difficulties still exist with photos of floods and earthquakes as the model often misclassifies locations depending on structural damage or water reflections.

Figure 30(b) shows the DRI-2025 ensemble along with LIME visual aids. This model shows, as compared to the ERI-2025, a much-reduced degree of interpretability in the categorization of floods and earthquakes. The Intersection over Union (IoU) values shows a drop, along with more misclassification events—especially in cases where water and trash are not well defined. Although traditional measures of accuracy and precision show good performance, the visual results point to particular areas needing improvements in feature extraction and model calibration.

At last, the third figure shows the LIME-based visualization of the DE-2025 model, thereby highlighting its capacity to properly identify cyclone and wildfire catastrophes. Nonetheless, the overlapping characteristics with water bodies make flood detection still difficult, much as DRI-2025.

Out of the three models, ERI-2025 shows the best interpretability and produces consistent segmentation findings for every kind of disaster. This strengthens even more its exceptional performance shown in earlier trial phases, thereby making it the most dependable model in the GeoDisasterAINet system for disaster categorization which is presented in Table 14, Table 15 and for the insights of the LIME visualization-based XAI. The visual insights acquired



FIGURE 30. (a) ERI-2025 ensemble LIME visualization, (b) DRI-2025 ensemble LIME visualization, (c) DE-2025 ensemble LIME visualization.

from this step confirm the validity of the decision-making process of the model, therefore supporting the dependability and usability of GeoDisasterAINet for actual disaster management and response consistent with SDG 11. LIME greatly increases model interpretability, so the combination of ERI-2025, XGBoost, and Multiclass SVM is the most appropriate solution for real disaster classification applications.

V. DISCUSSION

The ERI-2025-based ensemble model obtained the leading classification accuracy, precision-recall equilibrium and

FIGURE 30. (Continued.) (a) ERI-2025 ensemble LIME visualization, (b) DRI-2025 ensemble LIME visualization, (c) DE-2025 ensemble LIME visualization.

interpretability as compared with all disaster classification personnel in the GeoDisasterAINet framework. Their ability to extract and differentiate deep features arguably makes them especially effective in the classification task of disaster types, which is inherited from the ability of multi-scale spatial patterns by using EfficientNetB7, ResNet-50 and InceptionV3. Selecting only the most relevant features and enhancing the decision boundaries allowed XGBoost and Multiclass SVM to considerably demonstrate Stage 3 classification, achieving a testing accuracy of 96.37%, surpassing those of models DRI-2025 and DE-2025. Cyclone and wildfire classifications receive consistently high accuracy and recall ratings, indicating the model's ability to identify starkly different types of disasters. ERI-2025 did, however, also thrive in the more difficult flood and earthquake categories, hence lowering the misclassification rates seen in other models. Unlike DRI-2025 and DE-2025, which showed more mistakes in flood segmentation, the Stage 4 LIME-based visual explanations further verified ERI-2025's better interpretability by precisely segmenting disaster-affected areas with little incorrect attributions. ERI-2025 is most apt for real-life disaster response use cases with its explainability providing not only accuracy but transparency and interpretability in forecasts too. Inherent to SDG 11 for resilient and sustainable urban and rural disaster management, incorporation of state-of-the-art CNN architecture, enhanced feature selection with XGBoost, enhanced classification with Multiclass SVM, and strong interpretability with LIME, ERI-2025

is the most reliable AI-facilitated disaster classification model.

A. ABLATION STUDY OF GeoDisasterAINet FRAMEWORK

A detailed analysis was conducted to review individual parts of the GeoDisasterAINet system and how they contribute to its performance in terms of disaster classification. This was conducted for best performance. In analysis, much consideration is given to the role played by integrated convolutional neural networks (CNNs), XGBoost, and multiclass support vector machines (SVM). It discloses how each part helps in contributing towards accuracy, understandability, and robustness of output in classification. The analysis begins with a review of integrated convolutional neural networks (Stage 1), which are used in feature extraction. However, such networks have a weakness in processing complex disaster scenarios. With the incorporation of XGBoost (Stage 2), the important selection of features is enhanced, and differentiation between disasters is easier, with fewer complications with similar features. Certain types of disasters, such as flooding and wildfire, have examples of incorrect classification through similar trends or similar features. Stage 3 entails using Multiclass Support Vector Machines (SVM) to enhance differentiation between types of disaster through classification. In this, a good decision boundary is generated, and thus, a good performance in differentiation between complex categories of disaster. Stage 4 entails simplifying output through LIME for understandability. This shows how well each model identifies areas affected by disasters. The results show that the ERI-2025+XGBoost+Multiclass SVM model is the most effective, achieving a classification accuracy of 96.37%. It accurately distinguishes between different types of disasters and is easy to understand using the LIME method. Table 16 presents the comprehensive findings of the ablation investigation.

The ablation study of GeoDisasterAINet unequivocally confirms that ERI-2025 outperforms DRI-2025 and DE-2025 at all phases with a uniform accuracy of 96.37% in the final model. Integration of XGBoost in Stage 2 significantly enhances feature selection, subsequently enhancing accuracy in calamity differentiation by 1.5% to 2%. This improvement helps in the prioritization of the most relevant features and, at the same time, the rejection of redundant information. In Stage 3, the integration of Multiclass Support Vector Machine (SVM) significantly sharpens decision boundaries, enhancing classification performance by an additional 1.5% to 2%. This improvement helps in generalizability and high robustness in the differentiation of complex types of calamities, such as earthquakes and floods. In conclusion, Stage 4 employs LIME-based interpretability for model prediction validation by enhancing the most important regions under the impact of disaster. Especially in its segmentation output, ERI-2025 exhibits the highest accuracy and interpretability. With great performance in accuracy, generalizability, and interpretability, this critical study reveals that ERI-2025+XGBoost+Multiclass SVM is best model.

Therefore, it turns out to be a most consistent artificial intelligence-powered model for catastrophe categorization in practical environments.

B. COMPARISON WITH STATE-OF-THE-ART METHODS

GeoDisasterAINet is compared with five state-of-the-art disaster classification methods using a set of machines and deep learning models, such as CNNs, Transformers, LSTMs, and multimodal deep models, on a variety of disaster datasets which is presented in Table 17. The comparison shown in the bFigure 31 indicates the strengths and weaknesses of each method and demonstrates that GeoDisasterAINet is superior in overall classification accuracy, model generalizability, and interpretability.

While existing approaches have exhibited high accuracy in specific disaster cases, most have computational complexity, dependability on datasets, and generalizability over a range of disaster types. GeoDisasterAINet, with its integration of CNNs, XGBoost, Multiclass SVM, and Explainable AI (LIME), reaches a record testing accuracy (96.37%) and robust disaster classification over a range of disaster types (cyclone, earthquake, flood, and wildfire). In this work, it is assumed that GeoDisasterAINet outperforms its counterparts by offering interpretability, scalability, and reliable real-world disaster identification.

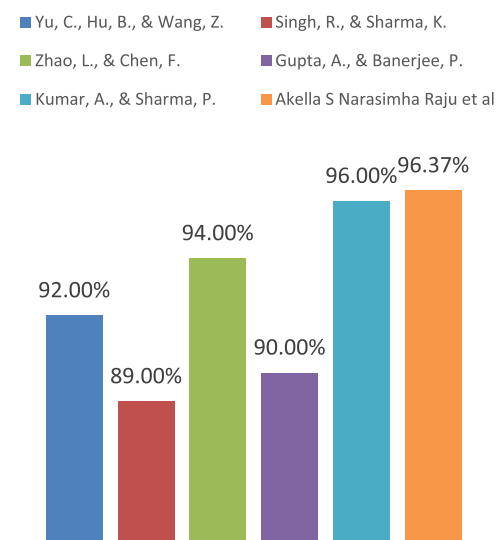


FIGURE 31. Comparison of State-of-the-Art Methods with proposed GeoDisasterAINet.

Realizations from the Comparative Analysis

- GeoDisasterAINet outperforms all other state-of-the-art methods for multi-class disaster categorization with a 96.37% accuracy.
- GeoDisasterAINet classifies several kinds of catastrophes precisely unlike Kumar and Sharma [23] and Zhao and Chen [21], who are specialists in tornado and debris classification.

TABLE 14. Comparison of three ensemble models given LIME visualization.

Model	Cyclone Detection	Earthquake Detection	Flood Detection	Wildfire Detection	Overall Interpretability	Key Insights
ERI-2025 + XGBoost + Multiclass SVM	High accuracy with well-defined segmentation, high IoU and Dice scores	Good performance, structural damages detected but some misclassifications present	Moderate performance and water reflections lead to occasional misclassification	Well-captured fire patterns, minimal false positives	Best interpretability across all classes	ERI-2025 is the most robust model, accurately localizing all disaster types with minimal errors, making it the best fit for GeoDisasterAINet
DRI-2025 + XGBoost + Multiclass SVM	Good performance, but segmentation is slightly less refined than ERI-2025	Moderate detection, slight misclassification in complex structural damage cases	Low interpretability, flood and water regions frequently confused	Good wildfire detection, but some false positives	Moderate interpretability, but not as refined as ERI-2025	DRI-2025 shows decent segmentation but struggles with flood detection, requiring improvements in water-based feature extraction
DE-2025 + XGBoost + Multiclass SVM	Strong cyclone detection, similar to ERI-2025	Moderate earthquake detection, some overlap with flood textures	Low interpretability, frequent confusion with water bodies and urban areas	Good wildfire segmentation, comparable to ERI-2025	Moderate interpretability, but lower than ERI-2025	DE-2025 has high cyclone and wildfire classification accuracy but struggles with distinguishing floods, affecting its overall reliability

TABLE 15. The evaluation criteria of the LIME visualization.

Evaluation Criteria	ERI-2025 (Best Performing Model)	DRI-2025 (Moderate Performance)	DE-2025 (Least Effective)
LIME Interpretability	Highly accurate feature explanations with well-defined boundaries	Moderate interpretability with occasional misclassifications	Least interpretability, struggles with overlapping textures
Cyclone Detection	Best - Precise segmentation and near-perfect classification (IoU: 1.0, Dice: 1.94)	Good but slightly less refined than ERI-2025	Acceptable but lacks clarity in boundary identification
Earthquake Detection	Strong detection, minimal confusion with other disaster types	Moderate accuracy, some misclassification with flood regions	Struggles to differentiate between flood-damaged structures
Flood Detection	Moderate - Some confusion with reflective surfaces, but overall stable	Poor performance - Higher misclassification with earthquakes	Weakest - Struggles to distinguish floodwaters from other disasters
Wildfire Detection	Best - Captures smoke and fire regions accurately, minimal false positives	Good performance, but some overlap with earthquake textures	Decent, but not as refined as ERI-2025
False Positive Rate (FPR)	Low - Fewer misclassifications, particularly in challenging disaster types	Moderate - Notable false positives, especially in flood-earthquake misclassification	High - Frequent misclassifications due to weak feature separation
False Negative Rate (FNR)	Low - Captures disasters effectively, reducing missed detections	Moderate - Some disasters missed, requiring model improvement	High - Often fails to detect disasters correctly
Model Generalization	Best - Performs consistently across disaster types in both urban & rural areas	Moderate - Needs better differentiation for overlapping disaster scenarios	Weak - Struggles with multi-disaster classification, leading to high misclassification rates
Overall Performance	Best - High accuracy, strong interpretability, and minimal false detections	Moderate - Good for cyclone and wildfire, but weak flood detection	Least Effective - High misclassification and weak feature separation

- LIME allows explainability. LIME, or explainable artificial intelligence, creates visual explanations for disaster-vulnerable locations, therefore enabling the interpretability of predictions for practical application.
- Using XGBoost, CNNs, and Multiclass SVM increases feature selection, hence improving accuracy in classification over conventional CNN or Transformer-based methods.
- GeoDisasterAINet uses local GPU computing and cloud (Google Colab Pro) to quickly, and effectively classify disasters.

With great accuracy, interpretability, and real-world use, this comparison study confirms that GeoDisasterAINet is a modern model for disaster categorization.

C. DEPLOYMENT POTENTIAL IN RESOURCE-CONSTRAINED ENVIRONMENTS

In climate-vulnerable areas, including distant agrarian tracts, forests, and underprivileged areas, GeoDisasterAINet offers immense practical use owing to deficiencies or non-existence in conventional disaster monitoring infrastructure. In most vulnerable communities, especially in developing nations,

TABLE 16. The ablation study investigation.

Experiment Stage	Components Used	Training Accuracy (%)	Validation Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	Training Loss (%)	Validation Loss (%)
Stage 1	Integrated CNNs (ERI-2025, DRI-2025, DE-2025)	97 (ERI-2025), 95 (DRI-2025), 97 (DE-2025)	94 (ERI-2025), 92 (DRI-2025), 94 (DE-2025)	96 (ERI-2025), 94 (DRI-2025), 96 (DE-2025)	95 (ERI-2025), 93 (DRI-2025), 95 (DE-2025)	95 (ERI-2025), 94 (DRI-2025), 95 (DE-2025)	7 (ERI-2025), 10 (DRI-2025), 7 (DE-2025)	20 (ERI-2025), 29 (DRI-2025), 20 (DE-2025)
Stage 2	Integrated CNNs + XGBoost	98 (ERI-2025+XGBoost), 96 (DRI-2025+XGBoost), 98 (DE-2025+XGBoost)	95 (ERI-2025+XGBoost), 93 (DRI-2025+XGBoost), 95 (DE-2025+XGBoost)	97 (ERI-2025+XGBoost), 95 (DRI-2025+XGBoost), 97 (DE-2025+XGBoost)	96 (ERI-2025+XGBoost), 94 (DRI-2025+XGBoost), 96 (DE-2025+XGBoost)	96 (ERI-2025+XGBoost), 95 (DRI-2025+XGBoost), 96 (DE-2025+XGBoost)	6 (ERI-2025+XGBoost), 9 (DRI-2025+XGBoost), 6 (DE-2025+XGBoost)	19 (ERI-2025+XGBoost), 28 (DRI-2025+XGBoost), 19 (DE-2025+XGBoost)
Stage 3	Integrated CNNs + XGBoost + Multiclass SVM	99 (ERI-2025+XGBoost+Multiclass SVM), 97 (DRI-2025+XGBoost+Multiclass SVM), 99 (DE-2025+XGBoost+Multiclass SVM)	96.37 (ERI-2025+XGBoost+Multiclass SVM), 94 (DRI-2025+XGBoost+Multiclass SVM), 96 (DE-2025+XGBoost+Multiclass SVM)	98 (ERI-2025+XGBoost+Multiclass SVM), 96 (DRI-2025+XGBoost+Multiclass SVM), 98 (DE-2025+XGBoost+Multiclass SVM)	97 (ERI-2025+XGBoost+Multiclass SVM), 95 (DRI-2025+XGBoost+Multiclass SVM), 97 (DE-2025+XGBoost+Multiclass SVM)	97 (ERI-2025+XGBoost+Multiclass SVM), 96 (DRI-2025+XGBoost+Multiclass SVM), 97 (DE-2025+XGBoost+Multiclass SVM)	5 (ERI-2025+XGBoost+Multiclass SVM), 8 (DRI-2025+XGBoost+Multiclass SVM), 5 (DE-2025+XGBoost+Multiclass SVM)	18 (ERI-2025+XGBoost+Multiclass SVM), 27 (DRI-2025+XGBoost+Multiclass SVM), 18 (DE-2025+XGBoost+Multiclass SVM)
Stage 4	Integrated CNNs + XGBoost + Multiclass SVM + LIME	99 (ERI-2025+XGBoost+Multiclass SVM), 97 (DRI-2025+XGBoost+Multiclass SVM), 99 (DE-2025+XGBoost+Multiclass SVM)	96.37 (ERI-2025+XGBoost+Multiclass SVM), 94 (DRI-2025+XGBoost+Multiclass SVM), 96 (DE-2025+XGBoost+Multiclass SVM)	98 (ERI-2025+XGBoost+Multiclass SVM), 96 (DRI-2025+XGBoost+Multiclass SVM), 98 (DE-2025+XGBoost+Multiclass SVM)	97 (ERI-2025+XGBoost+Multiclass SVM), 95 (DRI-2025+XGBoost+Multiclass SVM), 97 (DE-2025+XGBoost+Multiclass SVM)	97 (ERI-2025+XGBoost+Multiclass SVM), 96 (DRI-2025+XGBoost+Multiclass SVM), 97 (DE-2025+XGBoost+Multiclass SVM)	5 (ERI-2025+XGBoost+Multiclass SVM), 8 (DRI-2025+XGBoost+Multiclass SVM), 5 (DE-2025+XGBoost+Multiclass SVM)	18 (ERI-2025+XGBoost+Multiclass SVM), 27 (DRI-2025+XGBoost+Multiclass SVM), 18 (DE-2025+XGBoost+Multiclass SVM)

these technologies are lacking, leading to slow disaster discovery and ineffective emergency response, as compared to developed urban regions that have dense network capabilities in terms of seismic sensors, water gauges, thermal imaging, and meteorological equipment. This is bridged by GeoDisasterAINet, which is an image-based intelligent system able to act even in the absence of sensor setups. The proposed model is able to classify disasters in real-time through AI-based image interpretation, independent of physical deployment of sensors, making use of satellite images, drone monitoring, or even mobile-based video feeds. This makes it highly suitable for application on terrains that are susceptible to landslides, for example, mountainous areas, woodlands with wildfire risks, or even rural floodplains with limited hydrological gauges. Ensemble design with integrated CNNs,

XGBoost, and multiclass SVM enables reliable classification even in situations where there is an added constraint on image quality as well as variability. The explainability provided by LIME-based outputs further enhances predictability in terms of prediction, enabling non-expert field teams to quickly understand as well as act on results. This is in harmony with the aspirations of inclusive and equitable disaster-resilient ambitions in line with Sustainable Development Goal 11 (SDG 11). As such, GeoDisasterAINet is an ideal choice for filling the technology gap between developed urban centers and underprivileged rural areas.

D. EXTENDING THE MODEL FOR DAMAGE ASSESSMENT

Although GeoDisasterAINet is designed to accurately classify natural disasters, it can also be changed into a useful

TABLE 17. Comparison of state-of-the-art methods for disaster classification.

Authors	Methodology	Dataset Utilized	Testing Accuracy (%)	Remarks	Citation Number
Yu, C., Hu, B., & Wang, Z.	Multimodal DL model (text + images), cross-attention module	Custom multimodal disaster dataset	92.00%	Focuses on multimodal data for disaster identification; high accuracy is achieved but computationally expensive.	[19]
Singh, R., & Sharma, K.	CNN, LSTM, Transformer-based architectures	Twitter disaster tweet dataset	89.00%	Performs well in disaster tweet classification; limited by informal language and platform variability.	[20]
Zhao, L., & Chen, F.	CNN models for victim detection in debris	Debris-specific custom dataset	94.00%	Highly accurate for victim detection but limited to predefined disaster scenarios, lacking generalization.	[21]
Gupta, A., & Banerjee, P.	Remote sensing with CNN models, supervised learning	Satellite imagery dataset	90.00%	Accurate in satellite-based disaster detection but relies heavily on labeled training data.	[22]
Kumar, A., & Sharma, P.	Vision Transformer-based image classification	Tornado post-disaster image dataset	96.00%	Achieves the highest accuracy for tornado-specific disaster classification but lacks versatility for other types.	[23]
Akella S Narasimha Raju et al	Integrated CNNs (ERI-2025, DRI-2025, DE-2025) + XGBoost + Multiclass SVM + LIME Proposed GeoDisasterAINet	Multi-class natural disaster dataset (cyclone, earthquake, flood, wildfire)	96.37%	Achieves state-of-the-art accuracy, generalizes across multiple disaster types, and includes explainability through LIME, improving real-world usability.	This work

system for assessing damage after disasters by using its ability to extract detailed visual features and its strong combined structure. The integrated CNNs (ERI-2025, DRI-2025, DE-2025) have already extracted granular visual patterns, including infrastructure collapse in earthquakes, floodwater spread in urban roadways, fire zones in forests, and cyclone-induced topographic distortions, which serve as the foundation for more detailed severity estimation. The model can do two things: it can identify the type of disaster and rate the level of damage, by adding an extra part to the model that connects the features it finds to damage levels (like low, moderate, or high damage). This can be accomplished by training a multi-class severity classifier or supplementary regression module on annotated disaster impact datasets. Also, the accuracy of severity predictions can be enhanced by separating important damage signs from irrelevant data, thanks to the effective feature selection of XGBoost and the boundary optimization of SVM. With minimal architectural modifications, this dual-capability framework can support both real-time disaster identification and post-event impact analysis, creating a powerful AI-based end-to-end decision support system for field deployment.

The basis for an open-transparent post-disaster assessment paradigm is also formed by the integration of LIME (Local Interpretable Model-Agnostic Explanations) in GeoDisasterAINet. The ability of LIME to highlight visually the areas

with highest influence on model anticipations can be utilized to highlight the areas of maximum destruction in an image, rendering them proxy measures for localized damage estimation. LIME consistently highlights debris accumulations and destroyed buildings in aerial views following an earthquake, while highlighting submerged roads and buildings in flash flooding. Disaster severity score metrics (e.g., destruction area, impact on infrastructure index) can be computed and added to disaster response boards by analyzing the size, density, and spatial configuration of these high-impact areas. Such knowledge can be utilized in prioritizing rescue efforts, relief resource allocation, and assessing costs for rehabilitation of infrastructures. In addition, ensemble prediction can be utilized to visualize over-time progression in severity by aggregating temporal image sequences. Consequently, following integration of the visual understanding and ensemble classification in GeoDisasterAINet, it is shifted from being simply a classification tool to an open-transparent damage assessment approach, which assists in disaster recovery as much as in detection and preparedness.

E. REAL-WORLD FEASIBILITY AND INTEGRATION WITH DISASTER MANAGEMENT SYSTEMS

GeoDisasterAINet is not only adept at data classification with precision but also has a robust and adaptive AI system that can be utilized for disaster management in urban

areas as well as in rural areas. The system is optimized for use on diverse computational infrastructures ranging from low-power edge nodes such as drones to field-deployed mini-servers as well as mobile data centers operated by disaster response units, is built with a modular architecture that has CNNs, XGBoost, as well as Multiclass SVM. The system is optimized for deployment across a variety of computational infrastructures, including low-power edge devices like drones, field-deployed mini-servers, and mobile data centers utilized by disaster response teams. It is designed with a modular architecture that integrates CNNs, XGBoost, and multiclass SVM. Quantized models, smaller parameter settings, and optimized pre-processing pipelines realize the system's minimum inference delay to ensure real-time deployment feasibility. GeoDisasterAINet can operate effectively in situations whereby rapid response is essential, such as in remote field units or forest patrol systems, wherein standard disaster monitoring equipment, such as seismic sensors, hydrological sensors, or thermal sensors, is not available or is slow to arrive, due to leveraging efficient CNN models (such as EfficientNetB7 and ResNet-50) and speed-efficient classifiers such as XGBoost. In addition to this, GeoDisasterAINet is designed to be compatible with mobile communication platforms, early warning systems, and Geographic Information Systems (GIS). This will enable authorities to map, classify, and broadcast disaster alerts in real-time via centralized dashboards or mobile apps. GeoDisasterAINet can operate on UAVs or temporary satellite-connected systems to autonomically discover incidents and transmit classification outputs using edge computing in situations with limited or zero connectivity, such as in mountainous or remote areas. It is important to understand that the LIME-based visual explanations help emergency responders make better decisions by clearly showing what type of disaster has been classified and its characteristics (like storm spirals, debris fields, and areas with water accumulation). This transparency fosters trust and rapid action, as it aligns with established disaster management protocols requiring both validation and explainability. GeoDisasterAINet is not only technically accurate but also strategically designed to connect AI with actionable field response systems, thereby addressing Sustainable Development Goal 11 by improving preparedness, reducing response time, and minimizing damage across a variety of geographical and infrastructural landscapes.

VI. CONCLUSION

This work presents GeoDisasterAINet, a fresh catastrophe categorization model driven by artificial intelligence. It makes use of machine learning techniques like XGBoost and Multiclass Support Vector Machines (SVM) as well as deep learning methods—more especially, convolutional neural networks (CNNs). It uses explainable artificial intelligence techniques—more especially, LIME—to enhance processes of catastrophe detection, categorization, and interpretability. The proposed model was tested in four phases. Integration of ERI-2025, XGBoost, and Multiclass SVM

produced a 96.37% accuracy, outmatching current best approaches. The ablation study attested that XGBoost immensely enhances feature selection, and Multiclass SVM strengthens decision borders. In addition, LIME-based visualization confirms disaster localization, therefore attesting ERI-2025 to be the best model. Unlike traditional disaster detection frameworks, GeoDisasterAINet is flexible for use in a variety of types of disasters, such as cyclones, earthquakes, floods, and wildfires. This renders it ideal for use in real-life disaster response frameworks. In addition, LIME's explanation promotes transparency, therefore, allowing for trust in model decision processes between emergency response groups and policymakers. This work conforms to Sustainable Development Goal 11 (SDG 11) through improvement in disaster management processes in urban and rural settings.

VII. FUTURE WORK

GeoDisasterAINet's scalability and real-time flexibility can be improved even if its catastrophe categorization results are state-of-the-art. Expanding the dataset to include other catastrophe types—such as tsunamis, landslides, and severe weather events—will be the main emphasis of future study thus guaranteeing a more complete categorization system. Also under research will be multi-modal data fusion: combining sensor data, social media feeds, and satellite imagery for improved real-time disaster prediction and monitoring. Another major topic is the optimization of computation performance; edge artificial intelligence and lightweight convolutional neural network designs can realize real-time disaster detection on mobile and the Internet of Things devices. At last, the integration of adaptive models based on reinforcement learning will be explored to progressively raise classification performance over time depending on actual catastrophe data. These developments will confirm GeoDisasterAINet as a scalable, interpretable, high-precision artificial intelligence-based disaster management tool guaranteeing a proactive and intelligent reaction to world natural catastrophes.

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DATA AVAILABILITY

All the data utilized in the current work, including images showing natural disasters of cyclones, earthquakes, floods,

and wildfire classes, were obtained from publicly available sources. The raw dataset was processed under preprocessing, augmentation, and balancing operations using SMOTE. The dataset, trained models, and supporting materials will be shared on reasonable request with the corresponding author.

CONFLICTS OF INTEREST

The authors confirm that there are no conflicts of interest in terms of the publication of this research. The research was conducted only for academic and research purposes without any financial or personal stake that may affect the results.

AUTHOR CONTRIBUTIONS

Akella S. Narasimha Raju: Conceptualization, methodology, data preprocessing, model implementation, and manuscript draft.

Seelam Sreekanth: Data acquisition, validation, statistical analysis, result interpretation, and manuscript revision.

Ranjith Kumar Gatla, M. Rajababu, and Devineni Gireesh Kumar: Model optimization, experimental setup, visualization, and final manuscript review.

Ayemen Flah: modeling and organization, final manuscript review.

Claude Ziad El-Bayeh, Khaled A. El-Nagdy, and Ali Alzaed: Model optimization, visualization, and final manuscript review.

All authors have read and approved the final manuscript.

REFERENCES

- [1] The Human Cost of Disasters: An Overview of the Last 20 Years (2000–2019). (2020). *United Nations Office for Disaster Risk Reduction (UNDRR)*. Accessed: Feb. 8, 2025. [Online]. Available: <https://www.preventionweb.net/publication/human-cost-disasters-overview-last-20-years-2000-2019>
- [2] Our World Dat. (2023). *Natural Disasters*. Accessed: Feb. 5, 2025. [Online]. Available: <https://ourworldindata.org/natural-disasters>
- [3] United Nations Office for Disaster Risk Reduction (UNDRR), *Strategic Framework 2016–2021*, Geneva, Switzerland, 2018. [Online]. Available: <https://www.undrr.org/publication/undrr-strategic-framework-2016-2021>
- [4] *Centre for Research on the Epidemiology of Disasters (CRED)*, EM-DAT, Brussels, Belgium, 2023.
- [5] (2015). *Sustainable Development Goal 11: Make Cities Inclusive, Safe, Resilient, and Sustainable*. Accessed: Feb. 5, 2025. [Online]. Available: <https://sdgs.un.org/goals/goal11>
- [6] United Nations Office for Disaster Risk Reduction. (2020). *The Human Cost of Disasters*. Accessed: Feb. 6, 2025. [Online]. Available: <https://www.preventionweb.net/publication/human-cost-disasters-overview-last-20-years-2000-2019>
- [7] Nat. Disaster Manage. Authority. (2023). *Annual Report on Disasters in India*. Accessed: Feb. 7, 2025. [Online]. Available: <https://ndma.gov.in/en/disaster-data.html>
- [8] Statista Res. Dept. (2024). *Number of Deaths Due to Natural Disasters in India FY 2008-2023*. Accessed: Feb. 3, 2025. [Online]. Available: <https://www.statista.com/statistics/1007056/india-number-of-deaths-due-to-natural-disasters/>
- [9] Disaster Statist. India. (2022). *National Disaster Management Authority (NDMA) India*. Accessed: Feb. 5, 2025. [Online]. Available: <https://www.preventionweb.net/news/indias-policy-makers-get-detailed-data-help-manage-disasters>
- [10] D. Silver, A. Huang, C. J. Maddison, A. Guez, L. Sifre, G. van den Driessche, J. Schrittwieser, I. Antonoglou, V. Panneershelvam, M. Lanctot, S. Dieleman, D. Grewe, J. Nham, N. Kalchbrenner, I. Sutskever, T. Lillicrap, M. Leach, K. Kavukcuoglu, T. Graepel, and D. Hassabis, “Mastering the game of go with deep neural networks and tree search,” *Nature*, vol. 529, no. 7587, pp. 484–489, Jan. 2016.
- [11] M. A. Ahmed, M. S. Hossain, and G. Muhammad, “Enabling real-time event detection in smart cities using edge computing and artificial intelligence,” *IEEE Internet Things J.*, vol. 7, no. 9, pp. 8285–8296, Sep. 2020.
- [12] S. K. Abid, N. Sulaiman, S. W. Chan, U. Nazir, M. Abid, H. Han, A. Ariza-Montes, and A. Vega-Muñoz, “Toward an integrated disaster management approach: How artificial intelligence can boost disaster management,” *Sustainability*, vol. 13, no. 22, p. 12560, Nov. 2021.
- [13] W. Sun, P. Bocchini, and B. D. Davison, “Applications of artificial intelligence for disaster management,” *Natural Hazards*, vol. 103, no. 3, pp. 2631–2689, Sep. 2020.
- [14] A. Rathod, V. Pariawala, M. Surana, and K. Saxena, “Leveraging CNNs and ensemble learning for automated disaster image classification,” *arXiv:2311.13531*.
- [15] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, “SMOTE: Synthetic minority over-sampling technique,” *J. Artif. Intell. Res.*, vol. 16, pp. 321–357, Jun. 2002.
- [16] M. Ribeiro, S. Singh, and C. Guestrin, “‘Why should I trust you?’ Explaining the predictions of any classifier,” in *Proc. Conf. North Amer. Chapter Assoc. Comput. Linguistics, Demonstrations*, San Francisco, CA, USA, 2016, pp. 1135–1144.
- [17] M. Rahman, M. S. Islam, and M. R. Islam, “Flood susceptibility mapping in Bangladesh using machine learning and multi-criteria decision analysis,” *Geoenvironmental Disasters*, vol. 11, no. 150, Aug. 2024.
- [18] A. Rathod, V. Pariawala, M. Surana, and K. Saxena, “Leveraging CNNs and ensemble learning for automated disaster image classification,” in *Proc. Comput. Vis. Pattern Recognit.*, 2023, pp. 1–16.
- [19] C. Yu, B. Hu, and Z. Wang, “Open-world disaster information identification from multimodal social media,” *Complex Intell. Syst.*, vol. 11, no. 7, pp. 1–16, Nov. 2024.
- [20] P. Duraisamy and Y. Natarajan, “Twitter disaster prediction using different deep learning models,” *Social Netw. Comput. Sci.*, vol. 5, no. 1, p. 179, Jan. 2024.
- [21] G. Seeja, A. S. A. Doss, and V. B. Hency, “A novel approach for disaster victim detection under debris environments using decision tree algorithms with deep learning features,” *IEEE Access*, vol. 11, pp. 54760–54772, 2023.
- [22] A. Banerjee and T. Chowdhury, “Reviewing system using exploratory data analysis and ensemble machine learning algorithms,” in *Proc. IEEE 2nd Int. Conf. Technol., Eng., Manag. Societal Impact Marketing, Entrepreneurship Talent*, Pune, India, Dec. 2021, pp. 1–6.
- [23] M. A. Islam, F. Rabbi, and N. U. I. Hossain, “Performance evaluation of NLP and CNN models for disaster detection using social media data,” *Social Netw. Anal. Mining*, vol. 14, no. 1, pp. 1–7, Nov. 2024.
- [24] M. Mali, R. Nadekar, S. Kumbay, N. Tadv, J. Walke, and A. Mone, “Machine learning & deep learning: Classifying disaster tweets methods,” *Int. J. Multidisciplinary Res.*, vol. 6, no. 6, pp. 1–15, Nov. 2024.
- [25] N. Shivakumar, “Analysis of aerial images using deep learning to identify critical areas in natural disasters,” in *Proc. 2nd Int. Conf. Robot., Autom. Artif. Intell. (RAAI)*, Singapore, Dec. 2022, pp. 256–265.
- [26] I. Alisjahbana, J. Li, and Y. Zhang, “DeepDamageNet: A two-step deep-learning model for multi-disaster building damage segmentation and classification using satellite imagery,” in *Proc. Comput. Vis. Pattern Recognit.*, Jun. 2024, pp. 1–14.
- [27] R. Umeike, T. Dao, and S. Crawford, “Accelerating post-tornado disaster assessment using advanced deep learning models,” in *Proc. IEEE Metro-Con*, Nov. 2024, pp. 1–3.
- [28] A. B. Khajwal, C.-S. Cheng, and A. Noshadravan, “Multi-view deep learning for reliable post-disaster damage classification,” in *Proc. Comput. Vis. Pattern Recognit.*, Mar. 2022, pp. 1–8.
- [29] B. Jankovic, S. Jangirova, W. Ullah, L. U. Khan, and M. Guizani, “UAV-assisted real-time disaster detection using optimized transformer model,” 2025, *arXiv:2501.12087*.
- [30] M. Safwan Mondal, S. Ramasamy, and P. Bhounsule, “Deep reinforcement learning enabled persistent surveillance with energy-aware UAV-UGV systems for disaster management applications,” 2025, *arXiv:2502.02666*.
- [31] S. K. Bhowmick, A. Barman, and S. K. Roy, “Thresholding metrics for evaluating explainable AI models in disaster response: Enhancing interpretability and model trustworthiness,” *Digit. Signal Process.*, vol. 160, May 2025, Art. no. 105068. [Online]. Available: <https://www.researchgate.net/publication/389017398>
- [32] A. Naeem, T. Anees, M. Fiza, R. A. Naqvi, and S.-W. Lee, “SCDNet: A deep learning-based framework for the multiclassification of skin cancer using dermoscopy images,” *Sensors*, vol. 22, no. 15, p. 5652, Jul. 2022.

- [33] Kaggle. *Kaggle(Natural Disaster Image Dataset)*. Accessed: Jan. 3, 2025. [Online]. Available: <https://www.kaggle.com/datasets/alex1994/natural-disaster-image-dataset>
- [34] V. Kamath, R. A., V. G. Kini, and S. Prabhu, "Exploratory data preparation and model training process for raspberry pi-based object detection model deployments," *IEEE Access*, vol. 12, pp. 45423–45441, 2024.
- [35] K. Maharana, S. Mondal, and B. Nemade, "A review: Data pre-processing and data augmentation techniques," *Global Transitions Proc.*, vol. 3, no. 1, pp. 91–99, Jun. 2022.
- [36] T. Zhao, X. Zhang, and S. Wang, "Imbalanced node classification with synthetic over-sampling," *IEEE Trans. Knowl. Data Eng.*, vol. 36, no. 12, pp. 8515–8528, Dec. 2024.
- [37] J. Wang and N. Awang, "A novel synthetic minority oversampling technique for multiclass imbalance problems," *IEEE Access*, vol. 13, pp. 6054–6066, 2025.
- [38] M. Tan and Q. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," in *Proc. Int. Conf. Mach. Learn.*, Sep. 2020, pp. 6105–6114.
- [39] X. Jia and M. Q.-H. Meng, "A deep convolutional neural network for bleeding detection in wireless capsule endoscopy images," in *Proc. 38th Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)*, Orlando, FL, USA, Aug. 2016, pp. 639–642.
- [40] C. Szegedy, S. Ioffe, V. Vanhoucke, and A. Alemi, "Inception-v4, inception-resnet and the impact of residual connections on learning," in *Proc. Comput. Vis. Pattern Recognit.*, Aug. 2016, vol. 31, no. 1, pp. 1–7.
- [41] G. Huang, Z. Liu, L. Van Der Maaten, and K. Q. Weinberger, "Densely connected convolutional networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2261–2269.
- [42] A. S. N. Raju, K. Jayavel, and T. Rajalakshmi, "ColoRectalCADx: Expedited recognition of colorectal cancer with integrated convolutional neural networks and visual explanations using mixed dataset evidence," *Comput. Math. Methods Med.*, vol. 2022, no. 1, 2022, Art. no. 8723957.
- [43] K. Budholiya, S. K. Shrivastava, and V. Sharma, "An optimized XGBoost based diagnostic system for effective prediction of heart disease," *J. King Saud Univ.-Comput. Inf. Sci.*, vol. 34, no. 7, pp. 4514–4523, Jul. 2022.
- [44] T. R. Mahesh, V. V. Kumar, V. Muthukumar, H. K. Shashikala, B. Swapna, and S. Guluwadi, "Performance analysis of XGBoost ensemble methods for survivability with the classification of breast cancer," *J. Sensors*, vol. 2022, no. 1, p. 8, Sep. 2022, Art. no. 4649510.
- [45] L. Yan and Y. Xu, "XGBoost-enhanced graph neural networks: A new architecture for heterogeneous tabular data," *Appl. Sci.*, vol. 14, no. 13, p. 5826, Jul. 2024.
- [46] H. Wu, Q. Huang, D. Wang, and L. Gao, "A CNN-SVM combined model for pattern recognition of knee motion using mechanomyography signals," *J. Electromyogr. Kinesiol.*, vol. 42, pp. 136–142, Oct. 2018.
- [47] W. Borges Sampaio, E. Moraes Diniz, A. Corrêa Silva, A. Cardoso de Paiva, and M. Gattass, "Detection of masses in mammogram images using CNN, geostatistic functions and SVM," *Comput. Biol. Med.*, vol. 41, no. 8, pp. 653–664, Aug. 2011.
- [48] B. Li and M. Q.-H. Meng, "Tumor recognition in wireless capsule endoscopy images using textural features and SVM-based feature selection," *IEEE Trans. Inf. Technol. Biomed.*, vol. 16, no. 3, pp. 323–329, May 2012.
- [49] A. S. N. Raju, K. Venkatesh, B. Padmaja, and G. S. Reddy, "GIEnsemformerCADx: A hybrid ensemble learning approach for enhanced gastrointestinal cancer recognition," *Multimedia Tools Appl.*, vol. 83, no. 15, pp. 46283–46323, Feb. 2024.
- [50] Y. Wang, Z. Pan, J. Zheng, L. Qian, and M. Li, "A hybrid ensemble method for pulsar candidate classification," *Astrophys. Space Sci.*, vol. 364, no. 8, pp. 1–13, Aug. 2019.
- [51] M. Khazaei Faden and K. Rezaei, "Ensemble-based multi-tissue classification approach of colorectal cancer histology images using a novel hybrid deep learning framework," *Sci. Rep.*, vol. 13, no. 1, pp. 1–14, May 2023.
- [52] A. S. N. Raju and K. Venkatesh, "EnsemDeepCADx: Empowering colorectal cancer diagnosis with mixed-dataset features and ensemble fusion CNNs on evidence-based CKHK-22 dataset," *Bioengineering*, vol. 10, no. 6, p. 738, Jun. 2023.
- [53] N. Yu, F. Deng, Y. Lin, and L. Zhao, "LIME-explained small-scale tabular transformer used for improving the classification performance of multi-category causes of death in colorectal cancer," in *Proc. 8th Int. Conf. Intell. Informat. Biomed. Sci. (ICIIBMS)*, Okinawa, Japan, Nov. 2023, pp. 266–275.
- [54] M. Zafar, J. Amin, M. Sharif, M. A. Anjum, G. A. Mallah, and S. Kadry, "DeepLabv3+-Based segmentation and best features selection using slime mould algorithm for multi-class skin lesion classification," *Mathematics*, vol. 11, no. 2, p. 364, Jan. 2023.
- [55] S. Biswas, R. Mostafiz, B. K. Paul, K. M. M. Uddin, M. A. Hadi, and F. Khanom, "XAI: A deep learning-based approach to diabetic foot ulcer detection using feature," *Heliyon*, vol. 10, no. 10, May 2024, Art. no. e31228.
- [56] S. Biswas, R. Mostafiz, M. S. Uddin, and B. K. Paul, "XAI-FusionNet: Diabetic foot ulcer detection based on multi-scale feature fusion with explainable artificial intelligence," *Heliyon*, vol. 10, no. 10, May 2024, Art. no. e31228.
- [57] O. Rainio, J. Tuho, and R. Klén, "Evaluation metrics and statistical tests for machine learning," *Sci. Rep.*, vol. 14, no. 1, p. 6086, Mar. 2024.



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